Tier-D Ranks 21–40: Rank 29 Closure and N2 Checkpoint

Tactix engineering handoff

18 August 2026

Objective and current outcome

The active round is the frozen set of twenty Kissat-over-Tactix gaps at ranks 21–40 under the pre-Phase-4.5 contract: one solver thread, 16 GiB process limit, and 1,200-second solve limit. The objective remains to close all twenty with checked Tactix answers while preserving N1–N4, all 28 Tier-B instances, the fixed five Tier-C guards, and zero certificate residue.

This checkpoint closes Rank 29's first traced representation mismatches and produces a checked SAT solve in 130.767 seconds, versus the frozen eligible Kissat result of 294.468 seconds. The cumulative pin checks six of twenty ranks: 21, 23, 25, 28, 29, and 38. It is nevertheless rejected because the paired Tier-C guard found an N2 proof failure on `asconhashv12`. No performance promotion is claimed from a pin that fails correctness.

Rank 29: portable compaction-order repair

Before conflict 1000, Tactix and Kissat had the same directed binary order. Reduce-time variable compaction reconstructed both directions from one undirected stream and reversed the order. Preserving each directed prefix repaired that frontier, but the run still timed out. Continued common-state tracing exposed two later representation defects.

First, a replacement inherited its source's stable position but remained physically appended behind later clauses. Second, Reduce marked victims before sparse collection, while the collector ran only when root simplification changed some other clause. Thus pre-existing garbage could survive and the expected rewatch never happen. The accepted general repair collects for either pre-existing garbage or root rewriting, replays the swept tail by persistent source-position key, and stably partitions irredundant before redundant clauses. Tests use synthetic mixed tails and contain no benchmark, rank, literal, conflict, hash, or deadline identity.

Trace job `1634075` matches all 61,653 portable semantic records and all 297,047 normalized-watch records through conflict 2400. Runtime job `1634074` checks the resulting Rank-29 model. The code checkpoint is commit `9869a43`; the exact validation narrative is in `PROGRESS.md`.

Cumulative evidence and the blocking correctness result

Job `1634095`, in `benchmarks/runs/20260818-1930-tierd-ranks21-40-inherited-sparse/`, finishes all twenty tasks. Six answers validate (five SAT and one UNSAT), fourteen time out, peak RSS is 6,909,372 KiB, and certificate residue is empty. Ranks 23 and 29 are newly closed relative to the preceding cumulative pin, and no previously closed rank is lost.

Job `1634094`, in `benchmarks/runs/20260818-1930-BC-inherited-sparse/`, is 28/28 on Tier B and 4/5 on the fixed Tier-C subset. `asconhashv12` reports UNSAT in 101.224 seconds, but drat-trim

rejects the 15,605,586-byte certificate at the unit `-1997`. Every task still completes certificate cleanup, so residue is zero. The job manifest explicitly rejects the pin and requires soundness diagnosis before promotion.

First N2 hypothesis and focused falsification

The rejected proof contains hundreds of deletion lines for clauses the checker no longer holds. The common large-clause retirement primitive wrote a proof deletion before testing whether its arena record was already garbage. A second retirement was therefore an idempotent solver-state no-op but a second checker mutation. This is especially dangerous in DRAT: deletion is keyed by literals, so the surplus line can consume a different live duplicate and make a later unit cease to be RUP.

The first production repair checks the garbage bit before emitting the deletion, making the whole retirement transition idempotent. Sparse collection also drops a deferred OTFS reason if another rule already retired it; its dead `searched` word is a cursor, not a forwarding reference. Generic tests cover exactly one deletion for duplicate literals and safe collection of a retired deferred reason.

An optional diagnostic build feature shadows the checker's live clause population and aborts at the first unbacked deletion. It is armed only after the frontend has seeded the full input population, and production builds compile its emission hooks out. The rebuilt shadow suite passes 72/72; Release, Trace, and ASAN-O3 pass 72/72, 71/71, and 72/72.

Focused job `1634219` falsifies this as the complete Ascon cause. The production and shadow arms finish in 96.988 and 97.438 seconds and emit the same 15,605,586-byte certificate as the rejected predecessor; drat-trim again rejects unit `-1997`. The shadow does not abort. The warned surplus deletions therefore precede the point at which the frontend arms the oracle, during input finalization, or are absent from the seeded population. The production pin remains rejected. Both tasks remove all certificate scratch.

The extended oracle now seeds every raw DIMACS clause before normalization. It identifies the first real divergence on the generic `dupbinary.cnf` microcase: the parser stores normalized binary `-4 3` from raw clause `-4 3 3`, but DRAT had never been given the normalized multiset. Sparse collection later deletes `-4 3`; that deletion is unbacked even though the two clauses are set-equivalent. The prior assumption that DRAT required no input restatement was therefore false: additions use logical semantics, but deletions match the exact literal multiset.

The general repair queues narrowed stored clauses for both formats and adds their exact normalized form at `finish_input`. Native LRAT retains its explicit root-unit and original-id hints; DRAT lets the checker discover the same RUP justification. The shadow suite is now 72/72 clean, including all binary/ascii DRAT and native-LRAT microproofs. The repaired Release, Trace, and ASAN-O3 suites pass 72/72, 71/71, and 72/72. This is a local proof gate only; Ascon has not yet reverified, so the production pin remains rejected.

Focused job `1634231` then falsifies input restatement as the Ascon blocker. Both arms report 643 restatements, solve in 96.588 and 97.139 seconds, and emit the same 15,610,721-byte proof. Drat-trim still rejects unit `-1997`, now at proof line 768571 modulo deletion errors. The extended shadow does not abort, proving every emitted deletion has a backing clause in the checker's exact raw-input-derived population. All proof scratch is removed. The input repair is retained as a general N2 correction, but does not promote the production pin.

Forward-mode diagnostic job `1634233` verifies every preceding addition and proves that unit `-1997` itself is the first invalid inference, at checker clause id 1354312 and proof line 768571. Its neighboring additions are equivalence binaries, while solver statistics report 39 units from the V6a/Y1 binary-resolution channel. Congruence is therefore the leading emitter to isolate, though

this adjacency is not yet a causal proof. The 94.929-second solve and 55.770-second forward check leave no certificate residue.

Isolation job 1634236 makes the causal test: it disables only the existing V6a/Y1 channel. Ascon still solves UNSAT in 107.956 seconds, and drat-trim verifies the 63,635,534-byte proof in 60.224 seconds. The run reports zero Y1 findings/units and removes all certificate scratch. Thus Y1 is the necessary upstream channel on the bad trajectory. This is diagnostic evidence, not a promotion of the off arm: V6a remains retained by the T105/N4 result.

Premise-shadow job 1634243 then reaches the unchanged failure in 95.771 seconds without aborting. Both exact binary premises of every direct Y1 unit remain in the checker's population. Direct Y1 premise loss is therefore falsified: disabling Y1 changes the upstream root trajectory and avoids a bad unit emitted later within Congruence. The actual unit-emission channel still needs an origin label. Certificate residue is zero.

Unit-origin job 1634245 supplies that label: -1997 is a value-unit, emitted from Canon::Unit after value-derived gate canonicalization. It is not a direct Y1 unit. The run takes 95.686 seconds, reproduces the same certificate, and cleans all scratch. The next distinction is whether speculative RUP replay rejects the structural inference in the solver's own database or succeeds there and thereby proves a remaining solver/checker formula divergence.

A compile-time shadow-only diagnostic now forces that replay for every value-derived unit and prints its gate family plus pass/fail result. The Shadow and Release suites each pass 72/72 locally. The production build contains no runtime diagnostic branch, and no new production pin is promoted. The focused Ascon shadow task is campaign 20260818-2212-ascon-value-unit-replay, Slurm job 1634264, pinned from commit 1a53ba3 with shadow binary ef58cf2e.... Resume by reading its forced-rup label, checked result, and certificate cleanup state.

Job 1634264 completes the distinction. The bad unit is labeled value-unit -1997 kind=xor forced-rup=fail: it is not RUP even in Tactix's own clause database. The shadow drops that one unjustified inference, then solves UNSAT in 95.785 seconds; drat-trim verifies the 15,604,302-byte proof in 57.775 seconds. All certificate scratch is removed. This localizes a solver-side XOR Canon::Unit correctness defect, not checker-population divergence. The shadow's fail-closed replay is diagnostic only; the production repair must correct the canonicalization/root-value reasoning itself.

The next shadow refinement dumps the failed gate's output plus every original input, union-find representative, root value, and unit reason id. This will distinguish a stale extracted gate from incorrect parity/value folding before any production edit. The pinned campaign is Slurm job 1634270, from commit cdf3109 and shadow binary 79843e74...:

20260818-2220-ascon-xor-gate-dump

Job 1634270 identifies the exact shape: 1997=XOR(341,1562,1844), where 341=0 and 1562 is equivalent to 1844. The equivalent pair cancels, but the separate valued input made the implementation bypass the mandatory parity-elimination ladder. The repair routes every XOR with a nonempty cancelled set through that ladder, including mixed valued-plus-cancelled gates. A native-LRAT regression covers this generic shape. Release, Shadow, Trace, and ASAN-O3 pass 72/72, 72/72, 71/71, and 72/72 respectively. Focused production Ascon verification is campaign 20260818-2235-ascon-xor-ladder-repair, Slurm job 1634293, from commit ddc2ce4 and release pin 494b8bd....

Job 1634293 passes: production solves UNSAT in 95.680 seconds and drat-trim verifies the 15,610,817-byte proof in 57.220 seconds. Congruence reports eight units and zero unjustified inferences; all certificate scratch is removed. The focused blocker is cleared, so the paired 28B+5C

and frozen ranks 21–40 cumulative guards are next. No production pin is promoted until both complete.

Both guards are submitted from the exact focused-verified pin `494b8bd...`. The 28B+5C campaign is job `1634307`:

`20260818-2240-BC-xor-ladder`

The frozen ranks 21–40 campaign is job `1634294`:

`20260818-2240-tierd-ranks21-40-xor-ladder`

Neither array has a throttle or node constraint.

The 28B+5C array completes 33/33 valid: Tier B is 28/28 (14 SAT, 14 UNSAT), and Tier C is 5/5 (3 SAT, 2 UNSAT). Every answer is model/proof checked and certificate residue is zero. Ascon checks UNSAT again in 122.306 seconds plus 63.054 seconds verification. This accepts the paired correctness half; ranks 39–40 are the only Tier-D tasks still running.

The Tier-D array also completes cleanly. It closes the same six ranks as its predecessor—21, 23, 25, 28, 29, and 38—with no gains and no losses. All six answers check; the other fourteen tasks time out at 1,200 seconds. Peak RSS is 6,919,528 KiB and certificate residue is zero. The paired cumulative guards therefore accept release pin `494b8bd...`; the proof repair changes no Tier-D coverage. Rank 22 is the first still-open frozen target and becomes the next divergence diagnosis.

The initial paired rank-22 semantic trace is Slurm job `1634369`:

`20260818-2324-tierd-rank22-regression-trace`

It uses immutable Kissat/Tactix trace pins, 16 GiB, a 300-second solver limit, two unthrottled tasks, and no node constraint. Resume by comparing its portable streams with `tools/bench/compare\_event\_trace.py`.

Job `1634369` is not yet causal evidence. Kissat spends all 300 seconds emitting 309,592,064 bytes of watch detail inside early Lucky and never reaches the first shared macro boundary; Tactix used watch/imply/heap suppression and reaches search before its 306,472,422-byte trace truncates. Cleanup is clean, but the observer contracts differ. Both arms must be rerun with symmetric watch/imply/heap suppression before comparison.

The symmetric low-volume rerun is Slurm job `1634385`:

`20260818-2331-tierd-rank22-low-volume-trace`

It uses the same pins and resources. Both arms now suppress watch, implication, and heap detail. Both arms time out cleanly: Kissat at 300.129 seconds with a 262,586,368-byte trace, and Tactix at 300.133 seconds with a 306,472,422-byte truncated trace. Certificate residue is zero. Both reach search and their first two decisions agree.

The stock semantic comparison still stops before a causal claim, but now for observer-vocabulary reasons: Tactix suppresses the initial `top32` field while Kissat retains it, Tactix omits propagation-cursor records, and one pin reports binary minimization at a finer tick granularity. These differences do not establish a solver state divergence. Resume by normalizing or filtering those observer-only records, then compare decisions, conflicts, learned clauses, and macro boundaries to locate the first portable semantic mismatch.

After normalizing those records and Kissat's 4,095-byte per-record payload limit, every available decision and conflict agrees: 9,906 shared control records through conflict 527 and decision 9,389. Thus the apparent conflict-seven learned-clause mismatch was payload truncation, not solver divergence. The trace budget was instead consumed by 1,449,229 Tactix random draw records and by analysis ticks in both arms.

The next symmetric control trace is Slurm job `1634396`:

`20260818-2347-tierd-rank22-control-trace`

It suppresses random-draw and analysis-tick observations while retaining decisions, conflicts, and learned-clause stages. The immutable solver pins, 16 GiB limit, 300-second solver limit, unthrottled two-task array, and lack of a node constraint are unchanged. Trace trajectories run without proof mode; any emitted UNSAT answer is independently replayed with a proof and checked, and any SAT answer is model checked before it is accepted.

Job `1634396` completes with clean 300-second timeouts and zero certificate residue. Suppression reduces the stable traces to 9,576,448 bytes for Kissat and 19,008,305 bytes for Tactix; neither hits the record cap. The streams match for 10,051 decision/conflict records, including conflicts 532 and 533, then diverge at decision 9,529 on level 2,037: Kissat chooses external variable 3,859,653 while Tactix chooses 3,813,932.

This frontier is now causal enough to choose the next observer. Conflict 533, its four-literal learned clause, and its backtrack level are identical. Kissat's analysis-end snapshot names 3,813,932 as both the search candidate and heap top, yet its next decision is 3,859,653; Tactix selects the named candidate. The evidence therefore points to Kissat taking a random-decision branch that Tactix does not take, rather than to conflict analysis, clause minimization, score bumping, or heap order. Resume by tracing the random decision scheduler and draw acceptance immediately after conflict 533.

Trace-only commit `982fa9b` adds the missing Kissat random-draw observer; it reports the raw 32-bit value, modulo slot, external variable, and acceptance under the same conflict-window control already used by Tactix. The overlay materializes and builds cleanly from Kissat commit `8af8e56...`; the new Kissat diagnostic pin has SHA-256 prefix `54eef62c`. Focused Slurm job `1634399` is:

`20260819-0001-tierd-rank22-randec-window`

It observes only conflicts 532–533, suppresses watch, implication, heap, and analysis-tick detail, and caps each trace at 200,000 records. Both tasks remain unthrottled at 16 GiB with no node constraint. Compare the raw draw sequences, slots, external identities, and acceptance immediately before decision 9,529.

Job `1634399` completes cleanly and resolves the scheduler branch. Kissat emits 162,294 random-draw records in the window, while Tactix emits none. Immediately after conflict 533, Kissat rejects 1,419 draws and accepts raw value 2,423,939,830 at slot 3,859,651, mapping to external variable 3,859,653. Tactix's random sequence is inactive and it takes the ranked candidate. Therefore the mismatch precedes the LCG, modulo mapping, and acceptance test: it is in random-sequence start/countdown lifetime. Both traces time out cleanly with zero certificate residue and neither reaches its record cap.

The common history shows a focused restart after conflict 523, backtracking from level 286 to root, followed by the start of a random sequence. Resume by observing the sequence ordinal, computed length, remaining-conflict countdown, and next deadline at that start and after conflict 533. This will distinguish an ordinal/rounding mismatch from an off-by-one countdown mismatch before any production repair is attempted.

Trace-only commit `13cb7e1` adds that state observer to both solvers and marks it auxiliary in semantic comparison. Release passes 72/72 tests, Trace passes 71/71, and a freshly materialized Kissat overlay builds cleanly. Focused Slurm job `1634404` is:

`20260819-0014-tierd-rank22-randec-state`

It observes conflicts 520–533 while suppressing the high-volume draw stream. The 50,000-record cap, immutable pins, 16 GiB limit, unthrottled two-task array, no node constraint, independent answer checks, and cleanup contract are recorded in the manifest.

Job 1634404 completes as two clean 300-second UNKNOWN timeouts with zero certificate residue, but it is observer-invalid. Both solvers' existing SEARCH_ONLY allowlists filtered the new randec_state event, so the traces contain zero state records. The manifest records Kissat's 1,269,760-byte trace and Tactix's 1,412,735-byte trace, with both full SHA-256 identities. No production conclusion is permitted. Resume by adding the event symmetrically to both search/control allowlists, rebuilding and testing both trace binaries, then rerunning the identical frozen conflict-520–533 contract. That symmetric repair now builds cleanly in Tactix and a fresh Kissat overlay. The existing full Release 72/72 and Trace 71/71 suites pass; focused runtime/static regressions additionally require the state event to survive the compact filters. Commit 9fc6971 records that recovery point. The identical frozen-window rerun is now Slurm job 1634409:

20260819-0025-tierd-rank22-randec-state-rerun

Its manifest records fresh immutable Tactix/Kissat trace pins, the same conflict-520–533 window, suppressed random-draw detail, 50,000-record cap, 16 GiB and 300-second solver limits, unthrottled two-task array, no node constraint, independent answer replay/checking, and cleanup contract.

Job 1634409 fails immediately and safely as infrastructure-invalid: campaign materialization set every copied pin to mode 0444, so both tasks report PermissionError(13) before solver execution. No trace or answer was produced; both scratch directories were deleted and certificate residue is zero. Its manifest records the failure and forbids a production conclusion. The pin bytes were recopied into a new campaign, every declared SHA-256 and mode 0555 was validated, and both solver pins passed executable version smoke checks. Corrected Slurm job 1634412 is:

20260819-0028-tierd-rank22-randec-state-rerun2

All solver, trace, resource, answer-check, and cleanup contracts otherwise remain identical.

Job 1634412 completes validly as two expected 300-second UNKNOWN timeouts. Neither trace truncates; random-draw detail remains suppressed; Kissat emits 3,059 state records and Tactix 6,144. The trace identities are recorded in the manifest. Both tasks delete scratch state with zero certificate residue.

The state evidence resolves the cause. At conflict 523 both solvers start sequence 1, compute length 10, and set the next deadline to 1,023. At conflict 533 Kissat still reports remaining=1 and takes the random branch, while Tactix reports until=533, declares the sequence inactive, and takes the ranked candidate. The intervening conflict labeled 526 is produced during conflict analysis/OTFS, without a new decision or search-propagation call. Kissat decrements its counter only for conflicts returned by kissat_search_propagate; Tactix's absolute global-conflict deadline incorrectly charges the analysis conflict and expires one search-propagation conflict early. The production repair should replace that absolute lifetime with a remaining counter charged only by search-propagation conflicts, while preserving the already matching sequence ordinal, computed length, next deadline, LCG, slot mapping, and acceptance behavior.

That repair is now implemented without changing a proof rule. The branching subpolicy exposes a default no-op search-propagation-conflict hook; the default brancher decrements its remaining random-sequence counter there. Analyze's internal OTFS conflicts therefore advance the global solver statistics but not the random lifetime. A focused unit regression models one charged propagation conflict followed by an uncharged OTFS conflict, then checks random selection through exact counter expiry. Release passes 72/72 tests and Trace passes 71/71. The next step is a focused symmetric trace replay through conflict 533, followed by the rank-22 checked solver run and cumulative guards if the portable prefix now extends.

Commit a9f751c records the tested repair. Focused symmetric replay job 1634419 is:

`20260819-0040-tierd-rank22-randec-lifetime-replay`

It retains the frozen input and conflict-520–533 observer window, suppresses the draw stream, and uses freshly validated immutable pin hashes and executable modes. The 16 GiB/300-second unthrottled two-task contract, no node constraint, answer replay/checking, and residue cleanup remain unchanged.

Job `1634419` validates the focused repair. Both arms reach their expected 300-second UNKNOWN timeout, neither trace truncates, and both clean scratch state with zero certificate residue. At conflict 533 both report one remaining random-sequence conflict and decision 9,529 selects external variable 3,859,653. Projecting conflicts 523–533 onto decision/conflict records gives 2,908 records from each solver; normalized event type, conflict count, decision count, and payload are byte-identical. The generic whole-trace comparator still reports the already known startup representation/macro asymmetries and a final Kissat timeout-flush partial record, so this result closes only the declared bounded causal frontier. The manifest records both trace hashes and that scope. Proceed to a checked 1,200-second production run on frozen rank 22.

That production run is now Slurm job `1634423`:

`20260819-0049-tierd-rank22-randec-lifetime-runtime`

It pins the Release binary from repair commit `a9f751c`, the frozen UNSAT instance and hash, and independent `drat-trim`/model checkers. The single task has a 16 GiB process limit, 1,200-second solver limit, 4,800-second proof checker limit, no throttle or node constraint, and mandatory scratch/certificate cleanup before atomic result publication.

Job `1634423` does not close rank 22. The candidate reaches the declared 1,200.004-second limit without a final answer. It had written a 104,857,600-byte partial proof (SHA-256 recorded in the manifest), but UNKNOWN cannot be checked as UNSAT; the runner deletes that proof and all scratch state, leaving zero residue. Frozen Kissat remains checked UNSAT at 196.556 seconds. Thus the repair is accepted only as a focused parity correction: rank 22 stays open, the accepted release pin is unchanged, and divergence analysis resumes immediately after the now-matching conflict-533 frontier.

That continuation is Slurm job `1634432`:

`20260819-0112-tierd-rank22-post533-control-trace`

It freezes conflicts 534–560, suppresses random draws plus watch, implication, heap, and analysis-tick detail symmetrically, and raises the cap to 200,000 records so state observations cannot hide the next control mismatch. Both arms complete as valid UNKNOWN timeouts without truncation and delete all scratch state, leaving zero certificate residue. Each bounded projection contains 399 decisions and 18 conflicts; all 417 normalized event, conflict-count, decision-count, and payload records match byte-for-byte. The repaired portable prefix therefore extends through conflict 560. Rank 22 remains open; continue with a frozen window beginning at conflict 561, and authorize no production edit from an observer-only difference.

The frozen extension through conflict 1,000 is Slurm job `1634434`:

`20260819-0121-tierd-rank22-control-561-1000`

It reuses the exact immutable pins and symmetric suppression contract, with a 500,000-record cap sized for the wider state-event window. Both tasks complete as valid, untruncated UNKNOWN timeouts and delete all scratch state, leaving zero certificate residue. Each projection contains 13,753 decisions and 421 search-conflict records; all 14,174 normalized records match byte-for-byte through

global conflict 1,000. Combined with job `1634432`, the post-repair prefix has 14,591 matched control records after conflict 533. Rank 22 remains open; extend the bounded observer from conflict 1,001.

That extension is Slurm job `1634436`:

`20260819-0129-tierd-rank22-control-1001-3000`

It freezes conflicts 1,001–3,000 under the identical immutable pins and symmetric event suppression, raising the cap to 2,500,000 records for the wider window. Both arms publish valid UNKNOWN results without hitting the cap and delete all scratch state, leaving zero certificate residue. Their common projection matches through conflict 1,005, conflict 1,006 itself, and decisions 23,745–23,747. At stable-mode level four and global decision 23,748, Kissat selects external variable 28 while Tactix selects 10. Kissat's random state is inactive (remaining zero, next deadline 1,023, sequence one), and both solvers are stable, so this is an EVSIDS heap candidate/topology mismatch rather than another random-lifetime error. Resume with a narrow heap-score/top observer immediately before decision 23,748; no production edit is yet authorized.

That observer is Slurm job `1634438`:

`20260819-0138-tierd-rank22-heap-1005-1006`

It enables the existing symmetric stable-heap score/top events only for conflicts 1,005–1,006 while retaining suppression of watch, implication, analysis-tick, and random-draw detail. Both valid UNKNOWN arms are untruncated and clean all scratch. They learn the same unit `-10848` and share score bumps 5896, 5893, 5894, 10, 5931, 5928, 5930, and 5929. Kissat alone then bumps 28, 14, 15, and 70; both finish with 13785 and 13793. Consequently Kissat raises variable 28 from 2.1107952244 to 3.3385328876, while Tactix leaves it below variable 10 at 3.3107551098. This missing-bump set, not heap tie handling, causes decision 23,748. Resume by exposing variable 10848's assignment reason and nearby implication clauses in conflicts 1,005–1,006; no production edit is yet authorized.

That reason observer is Slurm job `1634440`:

`20260819-0147-tierd-rank22-reason10848`

It freezes conflicts 1,004–1,006, selects external variable 10848 for the assignment observer, and enables implication records while retaining all other low-volume suppressions. The immutable pins, one-million-record cap, 16 GiB/300-second unthrottled tasks, no node constraint, answer checks, and cleanup contract are recorded in the manifest. Both traces reach the cap, but the selected records occur before truncation. At conflict 1,005 both solvers assign `-10848` from binary reason `-3722682`; at conflict 1,006 both imply `10848` from the identical ternary reason `10848,13785,13793`. That direct reason accounts only for their shared final bumps 13785 and 13793. Kissat's extra 28, 14, 15, and 70 must therefore enter its main conflict-analysis set earlier in the traversal. The diagnostic Kissat arm emitted UNSAT in 230.121 seconds and replayed it in 196.292 seconds, but its 741,475,845-byte proof checker timed out at 600.104 seconds; that answer is explicitly invalid and the proof was deleted. Tactix timed out UNKNOWN at 300.504 seconds. Both arms leave zero certificate residue. Resume by comparing main-analysis admissions and binary/jumped reason traversal at conflict 1,006; no production edit is yet authorized.

That traversal observer is Slurm job `1634442`:

`20260819-0214-tierd-rank22-analysis1006`

It records every newly admitted main-analysis literal, resolved trail literal, stored binary endpoint, Tactix jumped-over endpoint, and nonbinary reason literal at conflict 1,006. The trace-only sources pass Release 72/72 and Trace 71/71; the Kissat overlay materializes and compiles. Immutable pins,

a 200,000-record cap, paired 16 GiB/300-second tasks, independent answer checks, and cleanup are predeclared. The accepted production source remains `a9f751c`. Both arms finish as valid, untruncated UNKNOWN timeouts and leave zero residue. Their traversals match through the first three nonbinary resolutions and three binary resolutions to endpoint `-10848`. The first difference is assigned literal 10: Kissat stores ternary reason `10,28,-14`, while Tactix stores jumped binary endpoint `-10848` over direct antecedent `3825735`. Kissat consequently enters the `28 -> 14 -> 15 -> 70` reason subgraph; Tactix skips it. This explains all four missing bumps exactly and exonerates bump/heap handling. Resume by localizing why propagation chooses those different reasons, beginning with trail positions and binary-watch order around `-10848`; no production edit is yet authorized.

The bounded implication follow-up is Slurm job `1634444`:

`20260819-0226-tierd-rank22-implications1006`

It enables the existing implication stream at conflict 1,006 only. Its 400,000-record cap avoids re-emitting preceding conflicts. Both focus prefixes are complete before later truncation and both tasks clean all scratch. Relative to 10848, both append 15 at +1, 70 at +926, -28 at +325716, and 14 at +326470 with identical reasons; -3825735 appears at +325723 in Kissat and +325726 in Tactix from the same binary. Kissat appends 10 immediately after 14 at +326471 from `-14,10,28`; Tactix appends it 72 entries later at +326542 with virtual reason `-10848,10`, whose direct antecedent is 3825735. Thus the owning divergence is propagation/watch pose around ternary `10,28,-14`. Kissat's traced UNSAT at 226.965 seconds is invalid: replay reproduces it, but the checker times out at 600.096 seconds. Its repeated 741,475,845-byte proof is deleted; Tactix is valid UNKNOWN at 300.496 seconds.

The filtered binary-origin journal is Slurm job `1634446`:

`20260819-0240-tierd-rank22-binary-journal`

It selects external variable 10 at root conflict zero, caps output at 5,000 transactions, and runs the validated 72/72 journal build for at most 180 seconds. Its target is derived pair `(10,3825735)`, absent from the input. The single 16 GiB task has no throttle or node constraint and cleans scratch; it times out cleanly with a 12-record, 3,099-byte journal and zero residue. No conflict-zero journaled transaction adds a binary involving variable 10. The pair is also absent both as an input binary and as two literals of any one original clause, so root transaction and direct root-shortening origins are excluded.

The first search-learning follow-up was Slurm job `1634447`:

`20260819-0245-tierd-rank22-analyze-journal`

It timed out cleanly in 300.319 seconds, left zero scratch residue, and wrote only the two 203-byte journal headers (SHA-256 `42ceb79f22961744d...`), with no transactions. This is a selector miss, not negative evidence: `Analyze` is not registered, while the atomic rule is `ConflictAnalysis`. Corrected Slurm job `1634448` preserved the validated binary, conflicts 1–1,005, external variable 10, 2,000-record cap, and single 16 GiB/300-second unconstrained task:

`20260819-0254-tierd-rank22-conflictanalysis-journal`

It timed out after 302.922 seconds with zero residue and an untruncated 51,215-byte journal (SHA-256 `3223c873b0c47a1b...`). The corrected rule selector emits 87 transactions through conflict 89. Its only variable-10 binary additions are `(-3826318,-10)` at conflict 5 and `(-10,3724608)` at conflict 64; target `(10,3825735)` is absent from that prefix. Full-formula pre/post digests and enormous

learned-clause serialization prevented coverage of conflicts 90–1,005, so later search learning remains unresolved.

The lower-overhead follow-up was Slurm job `1634449`:

`20260819-0302-tierd-rank22-watchpose1006`

Both arms return valid UNKNOWN timeouts, untruncated traces, and zero residue. Kissat writes 516,096 bytes (SHA-256 `114d19ed2fcae93c...`); Tactix writes 335,966 bytes (SHA-256 `f6d6d65f2d8385ef...`). Crucially, both conflict-1,006 `analysis_end` snapshots list 3825735 as a live binary partner of literal 10. Binary availability is therefore exonerated. The scans that assigned variable 10 occurred before conflict 1,006 was detected, while the counter still read 1,005, so the 1,006-only window did not capture the competing triggers.

Corrected Slurm job `1634451` expands only that boundary:

`20260819-0310-tierd-rank22-watchrace1005-1006`

Both independent arms again return valid UNKNOWN timeouts, untruncated traces, and zero residue. Kissat writes 581,632 bytes (SHA-256 `3d22734636698ea9...`); Tactix writes 397,078 bytes (SHA-256 `211a5132f6aaed9b...`). At conflict counter 1,005 and decision 23,744, Kissat scans ternary set `{-14,10,28}` in pose `(28,10,-14)`, with block 10, and then immediately scans the shared binary. The ternary therefore owns variable 10's reason. Tactix first scans the ternary in pose `(-14,28,10)`, with block −14; scans the binary 316 propagations later, assigning 10 from it; and only later scans and rotates the ternary to `(-14,10,28)`. The first causal mismatch is thus the derived ternary's pre-existing watched pose: Kissat watches `(28,10)`, while Tactix watches `(-14,28)` and leaves 10 unwatched.

Tactix clause id 13,469,364 inherits logical order 13,360,199. That input ordinal contains `-3825772`, `3820676`, and `-3820680`, not the target set, so the ternary is a substitution-derived image. Commit `2847d78` adds a disabled-by-default exact-clause-set pose probe to both trace implementations. It reports every matching live clause's physical literal order at macro snapshots and accepts a conflict ceiling; the rank-22 run limits its arena scan to conflict zero. Release passes 72/72, Trace passes 71/71, the production zero-overhead guard passes, and the materialized pinned Kissat overlay compiles cleanly. Slurm job `1634453` now runs the two independent 16 GiB/300-second arms from immutable pins:

`20260819-0327-tierd-rank22-clause-pose-origin`

Its exact target is `{-14,10,28}`; all detailed search records are disabled so the traces retain root preprocessing macro boundaries. Both arms return valid UNKNOWN timeouts with zero residue and untruncated traces. Kissat writes 466,944 bytes (SHA-256 `bb9180760e7f619c...`); Tactix writes 169,730 bytes (SHA-256 `8e7405cb03eb6668...`). Kissat first exposes the target after sweep in pose `(28,10,-14)`. Tactix has no corresponding inner macro snapshot, but its preprocessing-end record exposes clause id 13,469,364/order 13,360,199 in the same pose. Both late-lucky end snapshots then expose the same rotated pose `(10,-14,28)`. Clause creation and root preprocessing pose are therefore exonerated: search-time propagation introduces the watched-pose mismatch before conflict 1,006. Resume with exact-set `analysis_end` snapshots to bracket the first search conflict whose poses differ, then trace only that propagation interval.

The direct follow-up is Slurm job `1634455`:

`20260819-0338-tierd-rank22-watchhistory0-1005`

It reuses the immutable observer pins and records only watch scans of clauses containing variable 10 from root through conflict 1,005. The 500,000-record cap, 16 GiB/300-second allocation, independent answer checks, and cleanup contract are predeclared. Both arms reach the cap at conflict 248 because search-only mode still admits non-tick analysis records. Nevertheless, both traces contain exactly 63 target-clause watch records, and every scan literal, blocking literal, searched value, and physical pose matches through conflict 137. The Tactix arm is a valid UNKNOWN timeout with zero residue. Kissat reports UNSAT in 266.320 seconds and reproduces it in 191.930 seconds with a 741,475,845-byte proof (SHA-256 `473f644267a72758...`), but `drat-trim` times out after 600.097 seconds. That answer is unaccepted under the campaign contract. Both tasks remove their scratch and leave zero residue; the checker timeout is an unsolved answer, not evidence of a soundness defect.

Commit `77f3954` adds a disabled-by-default true watch-only gate: macro snapshots and watch records remain, while all nonsnapshot analysis traffic is discarded. Release passes 72/72, Trace passes 71/71, the production zero-overhead guard passes, and the materialized Kissat overlay compiles. Slurm job `1634457` resumes only the unproven conflict range 138–1,005:

```
20260819-0351-tierd-rank22-watchonly138-1005
```

Its 100,000-record cap is ample relative to the 3,832 watch records observed through conflict 248. Both arms return valid UNKNOWN timeouts with zero residue and untruncated traces. Kissat writes 1,572,864 bytes (SHA-256 `fa41c072cb40e0cc...`); Tactix writes 1,205,550 bytes (SHA-256 `c7fc442479821b2d...`). The last common target record is conflict 611, decision 11,410: both scan literal 14 with block 28 and pre-scan pose `(-14,28,10)`. Kissat's next target scan is at conflict 1,005, already in pose `(28,10,-14)`. Tactix instead scans at conflict 1,004 and 1,005 while retaining the old pose, then rotates after a second conflict-1,005 scan. The first latent pose divergence is therefore after the common conflict-611 scan and before Tactix's conflict-1,004 scan. Confirm the unobserved post-state with an exact-clause `analysis_end` snapshot at conflict 612; if it differs, record the three literal values at the conflict-611 scan.

Commit `8310f33` adds a lower conflict bound to the exact-clause observer; Release passes 72/72, Trace passes 71/71, the zero-overhead guard passes, and the materialized Kissat overlay compiles. Slurm job `1634459` now scans the clause arena only at conflict 612 and records its `analysis_end` pose:

```
20260819-0402-tierd-rank22-pose612
```

The two 16 GiB/300-second arms use immutable pins, a 10,000-record cap, full width, independent answer checks, and cleanup. Resume by comparing the two `clause_poses` fields; do not infer the post-scan pose from the pre-scan watch record alone.

Both conflict-612 `analysis_end` records are present and show the same pose `(-14,28,10)`. Tactix is a valid UNKNOWN timeout in 300.419 seconds. Kissat solves UNSAT in 294.474 seconds and reproduces that result in 189.569 seconds, but its independent DRAT check reaches the 600-second limit; the answer is therefore not accepted. Both traces are terminal and untruncated, and both tasks leave zero certificate residue. The common conflict-611 scan therefore does not create the latent mismatch.

The next mutation-capable boundary without another target watch scan is Kissat's conflict-1,000 reduction/collection, which Tactix does not enter. Slurm job `1634461` scans the exact clause only at that conflict and records the pre/post-reduce and shared switch snapshots:

```
20260819-0411-tierd-rank22-pose1000
```

It reuses the validated immutable pins. Each arm has 16 GiB and a 300-second solver limit, plus the usual answer-check and cleanup contract. Resume by finding the first conflict-1,000 snapshot whose pose differs.

Both arms return valid UNKNOWN timeouts with zero residue and untruncated traces. Kissat writes 532,480 bytes (SHA-256 `4977aecd1a5dfb4c...`); Tactix writes 443,628 bytes (SHA-256 `861876122d642578...`). Kissat retains pose `(-14,28,10)` at `analysis_end`, `reduce_begin`, and `reduce_tiers`; `reduce_end` changes it to `(28,10,-14)`, which persists through mode switch. Tactix emits no reduction at conflict 1,000 and retains `(-14,28,10)` through its switch. At the same boundary, Kissat's live redundant population falls from 679 to 661 and its binary count from 3,909,712 to 3,377,914; Tactix enters switch with 679 redundant and 3,909,628 binaries. The first causal pose divergence is therefore Kissat's conflict-1,000 reduction/collection boundary, absent in Tactix. Resume by comparing the first-reduction trigger and retained-clause collection/rewatch semantics against pinned Kissat, then implement the smallest source-backed parity repair.

Source comparison identifies the trigger defect. Tactix's `reset_schedule` chooses the 2,000-conflict arithmetic recurrence delta as the first deadline whenever the formula profile disables square-root cadence. Pinned Kissat initializes the first deadline unconditionally from `reduceinit=1000`; only *after* a completed reduction does its recurrence apply. Rank 22 selects Tactix's legacy arithmetic profile, so this mistake exactly explains the absent conflict-1,000 boundary. The retained clause path is already source-backed: `reduce_apply_refs` invokes the same sparse tail collection and rewatch operation. The candidate repair now always seeds the first deadline from `initial_interval` and adds a regression proving that an arithmetic recurrence cannot replace that initial deadline. The focused Release/Trace regression passes, followed by complete 72/72 Release and 71/71 Trace suites; the report rebuilds cleanly at 12 pages. The repair is committed as `c5a6928`. One-arm confirmation campaign

`20260819-0424-tierd-rank22-reduceinit1000`

runs as Slurm job `1634463` with 16 GiB, a 300-second solver limit, no array throttle or node constraint, and independent checking of any answer. It reuses the predecessor's immutable Kissat boundary. Accept only if Tactix now emits `reduce_begin`/`reduce_end` at conflict 1,000 and the target pose changes from `(-14,28,10)` to `(28,10,-14)` at the latter. Do not start production timing until this exact pose check passes.

Job `1634463` completes as a valid 300.368-second UNKNOWN timeout with zero residue and a terminal, untruncated 190,891-byte trace (SHA-256 `c8cfb576262127a...`). The schedule repair works: Tactix now emits `reduce_begin` and `reduce_end` at conflict 1,000 before switching mode. The pose acceptance fails, however: the target remains `(-14,28,10)` through `reduce_end`. Tactix's redundant population falls 679 to 652, whereas pinned Kissat falls 679 to 661. Its binary population falls 3,909,628 to 3,377,830; this is exactly Kissat's binary delta while preserving the pre-existing 84-entry offset. Thus scheduling and binary flush are repaired, but Tactix deletes nine extra redundant clauses and does not rewatch the irredundant target. Resume by comparing the conflict-1,000 `first_reducible` watermark and the retained/deleted clause set; determine why Kissat includes the target in the collected tail while Tactix preserves it as prefix. Production timing remains blocked.

The apparent watermark defect is a formula-profile defect one level above the collector. This exact 3,859,779-variable shape selects `LegacyStartupFactorOnly`, which correctly suppresses repeated search-time inprocessing but also inherited obsolete whole-search overrides: flat 75% reduction, static tiers, arithmetic cadence, and compaction off. Thus Tactix uses tiers `(2,6)` where Kissat's first Reduce uses `(3,6)`, deletes 27 rather than 18 clauses, and never admits Kissat's compacting whole-arena collection. The new candidate confines formula profiles to inprocessing, restoring every typed default search policy. It also computes `compactlim` from the union of retired and root-fixed variables, matching Kissat's inactive population; stable public identities remain distinct, but a due compaction still rewatches the whole arena and compacts the policy-owned random-index permutation. Focused kernel and strategy regressions pass, then the complete Release and Trace suites pass 72/72 and

71/71 respectively; the zero-overhead and documentation gates are included, and this report rebuilds cleanly at 12 pages. The candidate is committed as `9645e25`. Immutable one-arm campaign

`20260819-0437-tierd-rank22-profile-search-defaults`

runs as Slurm job `1634464`, again with 16 GiB, a 300-second solver limit, no throttle or node constraint, and independent checking of any answer. Accept only conflict-1,000 tiers `(3,6)`, exactly 18 redundant deletions, and target pose `(28,10,-14)` at `reduce_end`.

Job `1634464` passes all three checks. It is a valid 300.368-second UNKNOWN timeout with zero residue and a terminal, untruncated 358,395-byte trace (SHA-256 `951c2304eab80e5d...`). Tactix's conflict-940 tier snapshot is focused `(3,4)` and stable `(2,6)`, hence the intended Reduce hybrid `(3,6)`. At conflict 1,000 the redundant population changes 679 to 661, exactly Kissat's 18 deletions. Binary watches change 3,909,628 to 3,377,830, the same 531,798-entry drop as Kissat while preserving the known 84-entry pre-boundary offset. Most importantly, the target is `(-14,28,10)` through `reduce_begin`, becomes `(28,10,-14)` at `reduce_end`, and persists through both switch snapshots. The localized divergence is closed. Resume with one 1,200-second production timing arm under the normal independent answer-check and cleanup contract; do not run broad guards until that focused result is accepted.

The immutable production campaign is

`20260819-0445-tierd-rank22-profile-search-runtime`

with Release SHA-256 `39175031e7f7e729...` from repair commit `9645e25`. Slurm job `1634465` has 16 GiB, two CPUs, a 1,200-second solver timeout, no throttle or node constraint, and a 4,800-second checker allowance. SAT requires the pinned model checker; UNSAT requires the pinned DRAT checker; all certificate scratch is deleted before result publication. Resume by waiting for this job. Close rank 22 only for a checked answer; otherwise localize the earliest new divergence before broad guards.

Job `1634465` completes UNKNOWN at the 1,200.004-second cutoff, so rank 22 remains open and the accepted release pin is unchanged. The runner records an 80,740,352-byte partial certificate (SHA-256 `ae74cf31366460b...`) only as diagnostic evidence; no final answer was emitted, so it is not checked. All certificate and derived scratch is deleted, with `cleanup_pending=false`. Resume by comparing the already-captured post-conflict-1,000 macro traces and localizing their earliest new boundary; do not replay the closed admission/profile/compaction chain and do not start broad guards.

The saved traces localize that boundary without another run. At `reduce_begin(1000)`, both solvers' VMTF search cursor names external variable 3,824,785. Kissat's `compact_queue` maps and preserves this cursor through compacting collection, and its `reduce_end` snapshot still names 3,824,785. Tactix instead rebuilt its queue and unconditionally reset the cursor to the tail, changing the same field to 3,722,682 immediately at `reduce_end`; later retier and decision drift are downstream. The general repair maps the old cursor through variable compaction and falls back to the tail only if the cursor does not survive. Regression fixtures cover both real renumbering and Tactix's identity-preserving root-fixed compaction; Release passes 72/72 and Trace passes 71/71. Resume with one focused Slurm confirmation of conflict 1,000 before another production attempt.

That immutable confirmation is campaign

`20260819-0513-tierd-rank22-search-cursor`

at source commit `bcfe4a0`, with Trace binary SHA-256 `e7cecaabb2376d6b...`. Slurm job `1634466` has one task, 16 GiB, two CPUs, a 300-second solver limit, no throttle and no node constraint.

Resume by waiting for this job and checking that `search_external` remains 3,824,785 through `reduce_end(1000)` with a terminal, untruncated trace and zero certificate residue.

Job `1634466` confirms the repair. It is a valid UNKNOWN timeout in 300.419 seconds with zero residue and a terminal, untruncated 358,395-byte trace (SHA-256 `327f6b46979d30e...`). Tactix now keeps `search_external=3824785` at analysis end, Reduce begin/end, both mode switch snapshots, and both conflict-1,001 Rephase snapshots, exactly matching Kissat. The same aligned boundary exposes the next frontier: Kissat's live irredundant count changes 8,701,390 to 8,701,220 during compacting collection, whereas Tactix remains at 8,701,390. Localize those 170 root-satisfied clauses before another production timing run; do not replay the closed cursor arm.

Source inspection pins the mechanism to Kissat `reduce.c`: it samples `compactlim` before victim deletion and selects sparse-collection start as `compact ? 0 : first_reducible`. Tactix sampled compaction only after the Reduce rule had already collected from `first_reducible`, so its irredundant prefix could not be root-flushed. The general repair makes the pre-deletion admission explicit, passes it into Reduce, and reuses the same decision for variable compaction. A regression proves that compacting Reduce deletes a root-satisfied irredundant prefix that ordinary tail collection preserves. Release passes 72/72 and Trace passes 71/71; resume with one focused conflict-1,000 confirmation before any production run.

The immutable confirmation campaign is

`20260819-0524-tierd-rank22-compact-start0`

at source commit `115f40e`, with Trace binary SHA-256 `0ebef95861061041...`. Slurm job `1634467` has one task, 16 GiB, two CPUs, a 300-second solver limit, no throttle and no node constraint. Resume by waiting for this job and checking for irredundant 8,701,390 to 8,701,220 at `reduce_end(1000)`, while retaining the already-closed cursor, redundant population, target pose, terminal trace, and cleanup gates.

Job `1634467` closes this boundary. It is a valid UNKNOWN timeout in 300.369 seconds with zero residue and a terminal, untruncated 484,714-byte trace (SHA-256 `bdd9dc994ffa2075...`). At conflict 1,000 Tactix now matches Kissat on every targeted field: irredundant 8,701,390 to 8,701,220, redundant 679 to 661, binary 3,909,628 to 3,377,914, cursor 3,824,785, and target pose `(-14,28,10)` to `(28,10,-14)`. The next saved macro frontier is stable-mode retier timing: Tactix first retiers at conflict 1,254, versus Kissat at 1,293. Localize the decision-rate/retier clocks before another production run; do not replay the closed collection chain.

Because Kissat tests the retier deadline only on an internal OTFS conflict, the 1,254-versus-1,293 macro difference is downstream evidence, not yet the causal boundary. Campaign

`20260819-0532-tierd-rank22-post1001-search`

therefore compares pinned Kissat and Tactix decision/analysis records for conflicts 1,001–1,010 from the matched Rephase state. Slurm job `1634468` has two unthrottled tasks, each with 16 GiB, two CPUs, and a 300-second solver limit, with no node constraint. Resume by waiting for both tasks, validating terminal/untruncated traces and cleanup, then compare the portable decision, conflict, and learned streams for their first divergence.

Job `1634468` excludes an immediate search divergence. Both requested windows are complete and untruncated, and both tasks leave zero residue. Across conflicts 1,001–1,010 the solvers have identical 194 decisions, nine conflicts, 7,161 analysis admissions, 10,880 reason literals, 7,169 score bumps, and eight final learned clauses. At conflict 1,005 Tactix emits one extra raw resolution of external variable 10 and its jumped binary endpoint before both resolve 14,548; this is the known virtual jumped-reason representation. It changes no admission, bump, learned clause, or subsequent

decision. The next observer should suppress high-volume internal analysis records and extend the portable trajectory comparison from conflict 1,011 to the first retier boundary.

That compact observer is now implemented symmetrically and remains disabled unless the new trajectory-only environment toggle is set for the selected solver. Macro snapshots are retained, while detailed output is restricted to decisions, random/phase choices, conflicts, analysis transactions, and learned-clause stages. A focused regression proves that score bumps and analysis internals are suppressed without losing those portable records. Release passes 72/72 and Trace passes 71/71. The fresh observer binaries are pinned by SHA-256 prefixes `daf10cb6` (Tactix) and `38f0df0e` (Kissat), from source commit `f91a928`. Immutable campaign

`20260819-0547-tierd-rank22-trajectory-to1293`

runs as Slurm job `1634471`. Its two unthrottled tasks each have 16 GiB, two CPUs, a 300-second solver limit, and no node constraint; the trace cap is 100,000 records. Resume by waiting for both tasks, verifying trace completeness and zero residue, then comparing decisions, conflicts, analysis boundaries, and learned stages for the first mismatch.

Job `1634471` provides a valid partial localization. Both arms time out UNKNOWN with untruncated traces, independent result validation, and zero residue. The shared complete prefix through conflict 1,058 contains exactly matching 773 decisions, 38 conflicts, 38 analysis begins, and 38 analysis-end ranking snapshots. The Tactix arm then spends its remaining time formatting a very large learned clause at conflict 1,059, whereas Kissat's observer silently truncates learned payloads at 4,095 bytes. Consequently Tactix does not reach the requested retier boundary; this is an observer-cost/representation defect, not evidence of a solver trajectory or soundness defect. The next increment is to encode trajectory-only learned records as symmetric size, ordered-hash, and set-hash summaries, validate both implementations, and rerun conflicts 1,059–1,293. Do not invoke a solver repair or broad guard from this partial observer result.

The constant-size learned summary is now implemented symmetrically. In trajectory-only mode each learned stage records its clause size, a 64-bit hash of external literals in order, and a second hash after sorting the external literals; ordinary detailed tracing remains unchanged. The focused regression checks this encoding, Release passes 72/72, Trace passes 71/71, and both fresh Kissat and Tactix observer builds complete. Their SHA-256 prefixes are `4fe71d17` and `5e6ff832`, from source commit `9d27e72`. Immutable continuation campaign

`20260819-0559-tierd-rank22-trajectory-digest`

runs as Slurm job `1634473`. Its two unthrottled tasks each have 16 GiB, two CPUs, a 300-second solver limit, no node constraint, and conflict window 1,059–1,293. Resume by waiting for both tasks, verifying trace completeness and zero residue, then locating the first portable mismatch.

Job `1634473` validates the digest encoding and moves the causal frontier. Both arms are valid untruncated UNKNOWN timeouts with zero residue. Conflicts 1,059–1,073 have identical 34 decisions, 15 conflicts, 15 complete analysis transactions, and all four learned-stage digests. At conflict 1,074, the conflict and raw, sorted, shrunk, and final learned summaries also match; the final clause has size 1,343, ordered hash `ff7bac91198aff96`, and set hash `007f47a73518e914`. Tactix then times out before `analysis_end`, while Kissat continues through conflict 1,293. The first incomplete boundary is therefore after the policy emits the learned clause and before backjump completion. Resume by adding disabled stage markers around glue observation, chronological target selection, trail unwind, eager subsumption, assertion, and analyzed-variable bumping, then run one Tactix conflict-1,074 probe. This is a performance localization, not a soundness failure.

The narrow Tactix-only stage observer now brackets learned emission, policy clock updates, target choice, both trail unwinds, OTFS retirement, eager subsumption, assertion, and analyzed-variable

bumping. These records remain disabled outside event-trace builds and are retained by trajectory-only mode. The focused observer regression passes, followed by Release 72/72 and Trace 71/71; the fresh Trace binary has SHA-256 prefix `4459e81e`, from source commit `80a18b5`. Immutable campaign

`20260819-0609-tierd-rank22-backjump-stage`

runs as Slurm job `1634477`. Its one Tactix task has 16 GiB, two CPUs, a 300-second solver limit, no throttle or node constraint, and conflict window 1,074 only. Resume by waiting for the result, verifying the trace and zero residue, then reading the last emitted tail stage to identify the blocking sub-operation.

Job `1634477` disproves a post-learning stall. It is a valid 300.369-second UNKNOWN timeout with a terminal, untruncated 486,852-byte trace (SHA-256 prefix `69477411`) and zero residue. Every conflict-1,074 stage completes: target 465, both unwinds, OTFS retirement, eager subsumption, assertion, the 19,446-entry analyzed bump/reset, backjump return, and analysis end. The one-arm process then reaches conflict 8,756. The predecessor's missing analysis end was therefore a cutoff artifact: its two memory-heavy arms shared `node017` and Tactix reached conflict 1,074 only near 300 seconds. Resume with one Tactix-only trajectory window over conflicts 1,075–1,293, then compare it with the already-saved Kissat digest trace from job `1634473`. Do not treat co-scheduled diagnostic throughput as a solver regression.

The isolated continuation campaign is

`20260819-0618-tierd-rank22-tactix-to1293`

at source commit `c2fd235`, reusing immutable Trace SHA-256 prefix `4459e81e`. Slurm job `1634478` has one Tactix task, 16 GiB, two CPUs, a 300-second solver limit, no throttle or node constraint, and conflict window 1,075–1,293. Job `1634478` is a valid 310.089-second UNKNOWN timeout with an untruncated 126,412-byte trace (SHA-256 prefix `8142a529`) and zero certificate residue. It completes conflict 1,091 and stops during the conflict-1,092 analyzed-variable bump. Normalized comparison with the saved Kissat trace finds all 189 portable records for conflicts 1,075–1,091 exact: decisions, conflicts, analysis boundaries, and all four learned-clause digests. The conflict-1,092 common prefix also matches; Tactix lacks only `analysis_end` at cutoff. Source inspection identifies the dominant observer cost: trajectory mode made every analysis end scan the full multi-million-variable state and stably sort the active ranking for a snapshot; the scalar tail marker at cutoff is not causal. Trajectory-only analysis ends now remain portable records without full snapshots, and temporary `analysis_tail_*` records are filtered. The focused regression passes, followed by Release 72/72 and Trace 71/71. Resume by pinning this observer in an isolated continuation toward conflict 1,293 and comparing against job `1634473`. This is observer overhead, not a solver or soundness defect.

The compact observer is committed as `1002120`; its immutable Trace binary has SHA-256 prefix `34946c0b`. Continuation campaign

`20260819-0630-tierd-rank22-compact-to1293`

runs as Slurm job `1634479`. Its one full-width Tactix task has 16 GiB, two CPUs, a 300-second solver limit, no throttle or node constraint, and traces conflicts 1,092–1,293. Resume by waiting for this job, validating its result, trace completeness, and zero residue, then comparing portable records with the saved Kissat trace from job `1634473`. Ignore `top32` on compact analysis-end records; choices, conflict counters, levels, payloads, and all learned-clause digests remain directly comparable.

Job `1634479` is a valid 310.088-second UNKNOWN timeout with an untruncated 555,223-byte trace (SHA-256 prefix `bcb5cd6e`) and zero certificate residue. The compact observer completes

analysis through conflict 1,206, then records conflict 1,207 and all four learned-clause digests before cutoff. Resume by comparing conflicts 1,092–1,206 and the conflict-1,207 learned transaction with saved Kissat job `1634473`, ignoring `top32` only on compact analysis-end records. The normalized portable trajectory is exact from conflict 1,092 through conflict 1,198. Its first semantic mismatch is at conflict 1,199, decision 25,781, level 469: Tactix chooses external literal −3782027, while Kissat chooses −3782012. Both solvers first complete the conflict-1,199 learned transaction identically and choose −3779280 at decision 25,780; the mismatch is therefore after the conflict-1,199 analysis state. Preserve the exact prefix and discriminate score, queue/heap position, assignment, or phase before extending toward conflict 1,293.

Focused campaign

`20260819-0640-tierd-rank22-c1199-heap`

runs as Slurm job `1634482`. Its paired full-width search-only arms use the same immutable Kissat and Tactix observer pins, 16 GiB and two CPUs each, a 300-second solver limit, no throttle or node constraint, and detail conflict 1,199 only. Resume by waiting for both tasks, validating traces and cleanup, then comparing `heap_scores` immediately after analysis and after decision 25,780. The decisive evidence is each candidate's score and heap position just before decision 25,781.

Job `1634482` produces two terminal, untruncated, residue-free diagnostic traces. Kissat's 1,060,271-byte trace has SHA-256 prefix `0ab76385`; its initial UNSAT run takes 296.719 seconds and reproduces in 186.872 seconds with a 741,475,845-byte proof, but `drat-trim` reaches its 600.09-second limit, so the answer is not independently verified. Tactix's 898,260-byte trace has SHA-256 prefix `470bb5d9` and a valid 300.419-second UNKNOWN timeout. This campaign is diagnostic only and does not close rank 22. Resume by comparing the completed heap-score records at the two declared boundaries; do not treat the checker timeout as solver soundness evidence.

The heap comparison excludes score arithmetic, phase, and bump order. Both candidate variables have exactly score `25774.157807429845` in both solvers; both selected phases are negative; and all 38 conflict-1,199 score-bump variables occur in the same order. The remaining cause is heap pose among equal-score candidates: Kissat exposes 3,782,012 first, while Tactix exposes 3,782,027. Resume by adding a symmetric external-variable full-heap digest to diagnostic `heap_scores`, validating both observers, and locating the earliest pose divergence before conflict 1,199. Do not impose a global tie break without proving that it preserves the already-exact decisions.

The diagnostic observers now prefix `heap_scores` with the same FNV-style digest over the full external-variable heap pose. Tactix trajectory-only mode also returns before computing a heap record that its filter will discard. The focused regression passes, followed by Release 72/72 and Trace 71/71. Fresh observer SHA-256 prefixes are `3c6c29e0` for Tactix and `e92bf9a0` for Kissat. Resume with narrow conflict endpoints to locate the earliest digest divergence; ordinary solver behavior is unchanged.

Immutable survey campaign

`20260819-0706-tierd-rank22-heap-digest-survey`

runs as Slurm job `1634485`. Fourteen full-width tasks pair Kissat and Tactix at conflicts 1,000, 1,050, 1,100, 1,150, 1,175, 1,190, and 1,199. Each has 16 GiB, two CPUs, a 120-second solver limit, no throttle or node constraint, and an immutable observer pin. Resume by validating which endpoints were reached, then comparing each pair's final aligned post-analysis full-heap digest to bracket the first pose divergence. Record the bracket before adding any more detailed heap operation observer.

Job `1634485` is clean but non-comparable. All 14 tasks are valid UNKNOWN timeouts with untruncated traces and zero residue. Every Tactix arm reaches its requested endpoint, while the seven concurrent memory-heavy Kissat arms each reach only conflict 58 in 120 seconds. This is

an allocation-contention artifact and supplies no heap-pose bracket. Resume with one isolated Kissat conflict-1,000 task using the same pin, compare it against saved Tactix task 1, and advance adaptively only if the digest matches. Do not repeat the seven-arm Kissat concurrency pattern.

Isolated continuation campaign

`20260819-0711-tierd-rank22-kissat-c1000-heap`

runs as Slurm job `1634499`. Its sole Kissat task has 16 GiB, two CPUs, a 120-second solver limit, no throttle or node constraint, and detail conflict 1,000 only. Resume by validating endpoint coverage and comparing its aligned post-analysis digest with saved Tactix task 1 from job `1634485`.

Job `1634499` is a valid 120.117-second UNKNOWN timeout with an untruncated 16,384-byte trace (SHA-256 prefix `1ae775f7`) and zero residue, but it reaches only conflict 58. Thus the concurrent survey's miss was not solely contention; conflict 1,000 lies after a long prefix. Resume with one isolated 180-second Kissat endpoint, still below the frozen 196.556-second solve time, and require actual conflict-1,000 coverage before comparison.

The 180-second isolated continuation is campaign

`20260819-0715-tierd-rank22-kissat-c1000-heap180`

and Slurm job `1634500`. Resume by requiring actual conflict-1,000 coverage, validating cleanup, and comparing its aligned heap digest with saved Tactix task 1 from job `1634485`.

Job `1634505` validates the endpoint-only mechanism: it exits by the intentional SIGTERM after 26.379 seconds with a valid UNKNOWN result, zero residue, and an untruncated 3,322,340-byte trace (SHA-256 prefix `9ac5f48e`). It reaches conflict 1,000 and emits exactly one heap digest. That digest, `14650fb0739d0383`, is nevertheless not comparable: the analysis ends in focused mode, where Kissat's score heap is inactive. Saved Tactix task 1 emits its conflict-1,000 score-heap digest only after the immediate switch to stable mode (`4a915e9c2980d3a9`). Resume by moving the stop to the first stable `heap_scores` call at or after the selected conflict, rebuilding and testing, then rerunning conflict 1,000 for the aligned post-switch hash.

The pending-stop refinement records a focused analysis-end without stopping, suppresses the expensive mode-switch snapshots, and terminates only on the first subsequent stable heap callback; a stable analysis-end still emits its final digest directly, so intermediate score bumps cannot trigger the stop. The static regression and fresh optimized build pass; observer SHA-256 prefix is `5f098b86`. Resume with one isolated conflict-1,000 rerun.

The aligned continuation is campaign

`20260819-0752-tierd-rank22-kissat-c1000-stable`

and Slurm job `1634508`, again one 16 GiB/two-CPU task with no throttle or node constraint. Resume by requiring the focused conflict-1,000 `analysis_end` followed by exactly one stable `heap_scores`, then compare its ordered hash with `4a915e9c2980d3a9` from saved Tactix task 1.

Job `1634508` succeeds: intentional-SIGTERM UNKNOWN in 27.131 seconds, zero residue, and an untruncated 3,323,389-byte trace (SHA-256 prefix `beb151c4`). After the focused analysis-end, it emits exactly one stable heap digest. Kissat and Tactix hashes are both `4a915e9c2980d3a9`, and their top-32 score entries match. Thus the complete ordered stable heap pose is exact at conflict 1,000 and the divergence begins later. Resume with isolated conflict 1,050; saved Tactix task 3's final stable hash is `8bc36fc3f7d4250d`.

The next bracket point is campaign

`20260819-0755-tierd-rank22-kissat-c1050-stable`

and Slurm job `1634511`. Resume by requiring one final stable conflict-1,050 heap digest, an untruncated trace and zero residue, then compare with saved Tactix task 3 hash `8bc36fc3f7d4250d`.

Job `1634511` exits intentionally after 28.029 seconds with a valid UNKNOWN result, zero residue, and an untruncated 6,754-byte trace (SHA-256 prefix `1c116966`). Its final stable heap hash is `a5865471262a60e9`, differing from Tactix `8bc36fc3f7d4250d`. Yet both analysis-end ranked top-32 lists are exact. Thus score semantics remain aligned while only latent heap-array pose differs; together with exact conflict 1,000, the first pose change lies in conflicts 1,001–1,050. Resume with a paired conflict-1,025 endpoint and bisect by the ordered full-heap hash.

The paired midpoint is campaign

`20260819-0758-tierd-rank22-c1025-paired`

and Slurm job `1634516`, tasks 0–1. Both use immutable observer pins, 16 GiB and two CPUs with no throttle or node constraint; Kissat self-stops at the endpoint, while Tactix retains the validated 120-second diagnostic cutoff. Resume by validating both conflict-1,025 traces and cleanup, comparing final stable hashes, and halving the surviving interval.

Job `1634516` completes cleanly. Kissat self-stops in 30.179 seconds; Tactix reaches the endpoint and takes its valid 120.418-second diagnostic timeout. Both traces are untruncated and residue-free. Their final stable hashes differ: Kissat `d07ff09f458843c0`, Tactix `2550fac265c50b26`; their analysis-end ranked top-32 remains exact. The first latent heap-pose change is therefore in conflicts 1,001–1,025. Resume with paired conflict 1,012.

The next midpoint is campaign

`20260819-0803-tierd-rank22-c1012-paired`

and Slurm job `1634525`, tasks 0–1 under the same paired diagnostic contract. Resume by validating both conflict-1,012 traces and cleanup, comparing final stable hashes, and halving the surviving interval.

Job `1634525` completes with two valid, untruncated, residue-free UNKNOWN results. Kissat takes 30.230 seconds and Tactix 120.469 seconds. Anchoring each final heap record immediately before conflict-1,012 `analysis_end` gives Kissat `d35e14dd6e45ddf0` and Tactix `50040146ca81cd64`; later Tactix records at the same conflict count belong to subsequent decisions and are excluded. The learned transaction and ranked top-32 remain exact. The first pose change is in conflicts 1,001–1,012; resume with paired conflict 1,006.

The next midpoint is campaign

`20260819-0808-tierd-rank22-c1006-paired`

and Slurm job `1634528`, tasks 0–1 under the same paired contract. Resume by validating both conflict-1,006 endpoints and halving either 1,001–1,006 or 1,007–1,012 by the final stable hashes.

Job `1634528` completes with valid, untruncated, residue-free UNKNOWN results: Kissat 30.681 seconds, Tactix 120.369 seconds. Final conflict-1,006 hashes differ, Kissat `cb7647e239c08a03` versus Tactix `6ff40460694245dd`, while decisions, scores, and ranked top-32 remain exact. The first pose change is in conflicts 1,001–1,006. To avoid assuming that a pose mismatch cannot reconverge, resume with paired endpoints for every remaining conflict 1,001–1,005 in one campaign.

The exhaustive endpoint campaign is

`20260819-0812-tierd-rank22-c1001-1005-paired`

and Slurm job `1634531`, tasks 0–9. It pairs both immutable observers at each remaining conflict without a throttle or node constraint. Resume by validating all ten traces and cleanup, anchoring

the final heap record before each analysis-end, and naming the first conflict whose ordered hashes differ.

Job 1634531 is residue-free and every trace is untruncated, but only three new pairs are complete endpoints. Conflicts 1,003–1,005 all differ: Kissat/Tactix hashes are respectively 2cc932e10ca5ec55/ 830a0608aaffcf4f, c347670f0d35b6c8/ b50a01843bc30f92, and f630e2a89e515c73/ 6e2f172ab66acac9. Conflict 1,002 is not an analysis-end state: that transaction starts with counter 1,002 and ends with counter 1,003. Conflict 1,001 is a very large stable analysis; endpoint-only Kissat finishes with hash 915ba6872db85d6e, but Tactix's older observer hashes after every bump and times out before analysis-end. Thus the remaining question is whether 1,001 already differs or the 1,002-to-1,003 transaction creates the first mismatch. Resume by adding endpoint-only Tactix stop mode and rerunning only conflict 1,001.

Tactix now has the same disabled-by-default endpoint-only stop contract. Under TACTIX_EVENT_TRACE_STOP_AFTE it suppresses intermediate full-heap hashes, emits one final stable digest at analysis-end (or defers across a focused-to-stable switch), flushes, and terminates by SIGTERM. Ordinary runs are unchanged. The static regression, Trace 71/71, and Release 72/72 pass; fresh observer SHA-256 prefix is 68be1ac2. Resume with the isolated paired conflict-1,001 rerun.

The decisive endpoint rerun is campaign

20260819-0821-tierd-rank22-c1001-endpoint-pair

and Slurm job 1634542, tasks 0–1. Both endpoint-only observers use immutable pins, 16 GiB and two CPUs with no throttle or node constraint. Resume by requiring both conflict-1,001 analysis-end records and exactly one final heap digest each, then compare against Kissat 915ba6872db85d6e.

Job 1634542 resolves the boundary. Kissat self-terminates in 29.690 seconds; Tactix handles SIGTERM gracefully, emits its terminal record, and exits 0 in 84.813 seconds. Both results are valid, untruncated and residue-free. Their complete ordered heap hashes are exactly 915ba6872db85d6e, with matching top-32 scores. Therefore the first pose mismatch is created inside the next transaction, which begins with conflict counter 1,002 and ends with counter 1,003. Resume by recording aligned full-heap hashes before and after every score bump in that transaction and identifying the first operation with different pose evolution.

Both endpoint observers now support an additional disabled-by-default per-bump heap-digest mode for exactly that transaction. Pre-target snapshots remain suppressed; once the selected counter is reached, each score-bump callback emits the same external-variable ordered full-heap digest, followed by the ordinary final endpoint digest and stop. Static observer coverage, Trace 71/71, and Release 72/72 pass. The fresh Tactix and Kissat diagnostic SHA-256 prefixes are respectively 73123c8b and 623fd02f. Resume by launching the paired counter-1,002–1,003 capture under the existing 16 GiB, two-CPU, unthrottled diagnostic contract, then align each digest with its preceding bumped variable.

That capture is campaign

20260819-0831-tierd-rank22-c1002-bump-heaps

and Slurm job 1634545, tasks 0–1. Resume by waiting for both tasks, requiring valid untruncated traces and zero cleanup residue, then aligning each heap_scores digest with its immediately preceding score_bump.

Job 1634545 completes cleanly. Kissat self-stops in 30.479 seconds; Tactix exits 0 with a terminal record in 83.946 seconds. Both results are valid UNKNOWNs with untruncated traces and zero cleanup residue. Their seven bump variables and scores are exact. The first post-bump digest, after variable 3,820,024, already differs: Kissat 7a5db3480369f497 versus Tactix 9e7504332cc8c629; every later comparable digest also differs. Tactix records pre-bump digest b068116e76aa70af, but

the Kissat overlay lacks that symmetric checkpoint. This run therefore cannot yet distinguish a pose change during the intervening removals from one caused by the first bump. Resume by adding a per-bump-mode-only Kissat pre-bump digest and rerunning this same endpoint pair.

That symmetric Kissat checkpoint is now implemented only when the explicit per-bump endpoint mode is active; ordinary observer output is unchanged. The static overlay test, Trace 71/71, Release 72/72, and fresh optimized Kissat build pass. The new Kissat SHA-256 prefix is `a05d3072`; the unchanged Tactix pin remains `73123c8b`. Resume with the same two-task endpoint pair and compare the first digest before any score bump.

The rerun is campaign

`20260819-0840-tierd-rank22-c1002-prebump-pair`

and Slurm job `1634548`, tasks 0–1. Resume by waiting for both tasks, checking trace completeness and residue, then comparing their first `heap_scores` records before consuming any `score_bump`.

Job `1634548` completes cleanly: Kissat self-stops in 30.679 seconds and Tactix exits 0 with a terminal record in 84.003 seconds. Both valid UNKNOWN traces are untruncated and residue-free. The pre-bump heap already differs: Kissat `fa9ad0428dc1e7d9` versus Tactix `b068116e76aa70af`. Thus the seven score bumps are exonerated. Conflict 1,001 ended with exact heap `915ba6872db85d6e` at decision 23,721; the next conflict begins at decision 23,731. The first pose mismatch is created by one of those ten intervening decision/removal operations. Resume by hashing after decisions 23,722–23,731 while suppressing the large conflict-1,001 bump loop, then identify the first differing removal.

Both observers now accept a disabled decision-count threshold for intermediate heap digests while retaining the later conflict-count stop. Setting that threshold to decision 23,722 suppresses all hashes from conflict 1,001's large bump loop but retains the existing hash callback after each of the ten next decisions. Static coverage, Trace 71/71, Release 72/72, and the optimized Kissat build pass. Fresh Tactix/Kissat SHA-256 prefixes are `fd2143c6`/`7d6c8e21`. Resume with one paired capture from conflict 1,001 through the conflict-1,002 endpoint.

That capture is campaign

`20260819-0847-tierd-rank22-decision-heaps`

and Slurm job `1634551`, tasks 0–1. Resume by waiting for both tasks, validating trace completeness and cleanup, then pairing each decision literal with the immediately following full-heap digest to find the first mismatch.

Job `1634551` captures all ten required decisions in both untruncated, residue-free traces. Kissat reaches the 120.127-second ceiling only after that coverage; Tactix self-stops in 87.312 seconds. The very first decision, 23,722, selects literal `-3826966` in both solvers from the exact `915ba6872db85d6e` starting heap. Immediately afterward, however, Kissat's heap is `0c02c94ab22cef2d` and Tactix's is `3f0b92d9fb905b7d`. This originally suggested that the first decision's lazy heap-pop/removal path created the mismatch; job `1634554` below disproves that interpretation. The relevant mutation is the preceding level-551-to-root backtrack and heap reinsertion stream.

The targeted path observer now records each lazy pop's removed maximum and last element, every selected child move with both exact scores and positions, and a full digest after each completed pop. A heap-pop-only filter suppresses all unrelated analysis detail. Both controls are disabled by default. Static coverage, Trace 71/71, Release 72/72, and the optimized Kissat build pass; fresh Tactix/Kissat SHA-256 prefixes are `1cd42ec0`/`8ce35871`. Resume with one paired capture targeting Kissat decision 23,722 and Tactix's pre-increment count 23,721.

That capture is campaign

`20260819-0857-tierd-rank22-first-pop-path`

and Slurm job `1634554`, tasks 0–1. Both tasks are valid UNKNOWN runs, untruncated and residue-free. Kissat reaches its 120.117-second limit with an empty filtered trace; Tactix self-stops in 86.691 seconds with an 11,895-byte trace (SHA-256 prefix `e58ce3d1`). Neither trace contains a `heap_pop`. Reading this result against the event sequence resolves the apparent paradox: after conflict 1,001 both solvers backtrack from level 551 to root and reinsert newly unassigned variables; the resulting root is already unassigned, so selecting decision 23,722 does not pop it. Thus the first-pop hypothesis is disproved. Resume by comparing a bounded count and rolling digest of the c1,001 heap-reinsertion stream, then instrument only the first differing insertion path if the external reinsertion sequence itself agrees.

That comparison is campaign

`20260819-0905-tierd-rank22-reinsert-sequence`

and Slurm job `1634558`, tasks 0–1; scheduler dry run `1634557` passed. It reuses the pinned `8ce35871`/`1cd42ec0` observers, requests two CPUs and 16 GiB independently per task with no throttle or node constraint, stops at conflict 1,002, and caps each control-only trace at 100,000 records. Resume by validating both UNKNOWN outcomes, trace truncation and scratch cleanup, then extracting the conflict-1,001 `heap_reinsert` payloads in order. The first unequal identity localizes the trail-order divergence; complete equality instead calls for a per-insertion heap-pose checkpoint.

Both tasks are valid, untruncated, and residue-free UNKNOWN runs. Kissat reaches the 120.133-second limit after all required coverage with an 11,034,624-byte trace (SHA-256 prefix `8dee55ad`); Tactix self-stops terminally in 87.326 seconds with 10,260,808 bytes (prefix `555fa94d`). Both c1,001 streams contain exactly 32,690 variables and their sorted payloads have the same `28b4db00` digest, proving equal membership. Their order digests differ, however: Kissat `43c83964`, Tactix `7f847d91`. Items 1–42 agree; item 43 is `3858960` in Kissat and `3858955` in Tactix, followed by two permutations of the same small blocks. This trail-order difference is the first demonstrated cause of the heap-pose split. Resume by querying those two variables' true levels, trail positions, and reasons at conflict 1,001, then localize the earlier assignment-order divergence.

That four-way query is campaign

`20260819-0914-tierd-rank22-assignment-pair`

and Slurm job `1634561`, tasks 0–3; scheduler dry run `1634560` passed. Each solver/variable pair is independent at two CPUs and 16 GiB, with no throttle or node constraint, and stops at conflict 1,001. Resume by validating the four outcomes and cleanup, then align each selected variable's `assignment` record (true level, trail index, binary flag, and reason).

All four runs are valid, untruncated, and residue-free UNKNOWN outcomes. Kissat self-terminates on the requested signal in 33.899/33.194 seconds; Tactix emits terminal records and exits zero in 89.872/89.866 seconds. Both variables are level-1 binary implications of the same external reason literal `-3722682` in both solvers. Kissat places `3858960` at trail position 334,399 before `3858955` at 334,630; Tactix reverses them, placing `3858955` at 583,800 before `3858960` at 583,809 (both Tactix assignments are marked jumped). Therefore the reinsertion mismatch inherits a binary-propagation ordering difference, not a different implication reason or level. Resume by capturing the ordered binary watch list for `-3722682` at conflict 1,001 and locating the two consequents within it.

That paired snapshot is campaign

`20260819-0919-tierd-rank22-binary-watch-order`

and Slurm job 1634566, tasks 0–1; scheduler dry run 1634565 passed. The existing disabled-by-default annotation externalizes binary watch partners in stored order. Each task uses two CPUs and 16 GiB independently, with no throttle or node constraint, and stops at conflict 1,001. Resume by validating both outcomes and cleanup, then compare the annotated lists and the positions of 3858960 and 3858955.

Both tasks are valid, untruncated, and residue-free UNKNOWN runs. Kissat exits on the requested signal in 29.933 seconds with 6,625,794 trace bytes (SHA-256 prefix b29b9263); Tactix emits a terminal record and exits zero in 84.812 seconds with 6,529,678 bytes (prefix e72f479b). Both ordered binary annotations are exactly -15,3722637,-3722690,-306062; neither contains 3858960 or 3858955. Source inspection resolves this negative result: binary reason jumping replaces the immediate antecedent with its ancestor, and -3722682 is that stored virtual reason. Its equal watch list is exonerated. Resume by adding a disabled selected-consequence observer at binary assignment time that records the immediate other-false literal, whether reason jumping fired, and the final stored reason in both solvers.

That observer is now implemented symmetrically and remains disabled unless a selected external consequence is configured. It records the assigned literal, the immediate other-false literal seen by propagation, whether reason jumping fired, and the final stored reason. Static overlay coverage, Trace 71/71, Release 72/72, and a fresh optimized Kissat build pass. The fresh Tactix/Kissat SHA-256 prefixes are 7f18c827/dab54f97. Resume by pinning those binaries and running the two selected variables against both solvers through conflict 1,001.

That four-way capture is campaign

20260819-0929-tierd-rank22-binary-immediate

and Slurm job 1634569, tasks 0–3; scheduler dry run 1634568 passed. Each task uses two CPUs and 16 GiB independently, with no throttle or node constraint, and stops at conflict 1,001. All four outcomes are valid UNKNOWN; all traces are untruncated and residue-free, and cleanup completed. The two per-variable record counts agree exactly across solvers (123 and 134). For both variables, the final assignment occurs at conflict 611, decision 11,410, level 1, with immediate antecedent -28, reason jumping enabled, and stored reason -3722682. Their recent semantic event sequences also match. Thus neither immediate-reason construction nor jumping explains the reversed trail order: it lies in ordered scanning of -28's binary watch list. Resume by adding a compact disabled observer that reports the complete list count/hash and the positions of 3858960 and 3858955 at conflict 611.

That observer is now implemented symmetrically under the common SAT_EVENT_TRACE_WATCH_TARGETS control. When selected, it suppresses the unbounded literal rendering and emits only the binary count, a full ordered hash, and each requested target's one-based position plus signed literal. A generic two-entry fixture pins ordering, sign, and compactness. Trace passes 71/71, Release passes 72/72, static overlay coverage passes, and fresh optimized Kissat construction passes. The fresh Tactix/Kissat SHA-256 prefixes are 63445c15/8933faa7. Resume by pinning these binaries and running a paired conflict-611 snapshot for watch literal -28 and targets 3858960, 3858955.

That paired capture is campaign

20260819-0942-tierd-rank22-watch-positions

and Slurm job 1634574, tasks 0–1; scheduler dry run 1634573 passed. Each independent task has two CPUs and 16 GiB, with no throttle or node constraint, and stops at conflict 611. Resume by waiting for both tasks, then validate answer, truncation, and cleanup before comparing binary_count, binary_hash, and both signed one-based target positions.

Both tasks completed as valid UNKNOWN resource timeouts with untruncated traces, no residue, and completed cleanup. Their conflict-611 snapshots are decisive: both -28 lists contain 261

binaries, but Kissat's ordered hash prefix is `8d014614`, with `-3858960` at position 9 and `-3858955` at 246; Tactix's is `bf9f3d21`, with the same literals at 248 and 235. This directly explains the reversed propagation and trail order. The diagnostic also exposed a focused-mode observer defect: the requested stop marked a pending stop but never raised the signal, so both arms continued to the 120-second resource bound. The requested snapshots remain valid. Resume by repairing that observer-only stop contract, then capture the same compact summaries at preprocessing/rule boundaries to locate where the 261-entry order first splits.

The symmetric observer stop contract is repaired: every selected `analysis_end` now flushes and raises `SIGTERM` immediately in both search modes, after the optional stable heap snapshot. The unreachable pending state and its repeated focused snapshots are removed. A forked runtime unit test requires focused mode to die by the selected signal. Trace passes 71/71, Release passes 72/72, static overlay coverage passes, and fresh optimized Kissat construction passes. The new Tactix/Kissat SHA-256 prefixes are `bfcd50f0`/`4652be26`. Resume with the preprocessing-boundary capture; the diagnostic repair does not change solver behavior or the job-1634574 finding.

The boundary capture is campaign

`20260819-0952-tierd-rank22-watch-boundaries`

and Slurm job `1634577`, tasks 0–1; scheduler dry run `1634576` passed. Each task has two CPUs and 16 GiB independently, with no throttle or node constraint. It deliberately omits a conflict stop so startup and preprocessing snapshots remain visible, while detailed search is restricted to conflict zero. Resume by validating both outcomes and cleanup, aligning common boundary names, and identifying the first differing count/hash or target pose.

Both tasks are valid UNKNOWN runs with untruncated traces, no residue, and completed cleanup. At Sweep begin, both solvers have exactly six `-28` binary partners, ordered hash `a1cf11345e8485b0`, and neither target. Sweep creates the first split: Kissat ends it with 264 entries, hash prefix `9db104b9`, and target positions 11/251; Tactix ends it with 260 entries, hash prefix `7ff52c81`, and positions 248/235. Substitute then leaves 261 entries in both but preserves the prior order split. Resume by adding a disabled selected-literal binary-add observer at the symmetric new-binary insertion points and compare every runtime addition involving `-28` through preprocessing.

That selected binary-add observer is now implemented at the symmetric actual insertion points. Under `SAT_EVENT_TRACE_BINARY_ADD=-28`, it emits only new binary clauses containing that exact signed literal and is otherwise erased/disabled. A runtime fixture checks signed pair payloads; static coverage pins both insertion hooks. Trace passes 71/71, Release passes 72/72 in an isolated full run, and fresh optimized Kissat construction passes. The new Tactix/Kissat SHA-256 prefixes are `fa186938`/`b65aecfe`. Resume by pinning these binaries and capturing the conflict-zero preprocessing addition sequence for `-28` in both solvers.

That paired add-sequence capture is campaign

`20260819-1002-tierd-rank22-binary-adds`

and Slurm job `1634580`, tasks 0–1; scheduler dry run `1634579` passed. Each task has two CPUs and 16 GiB independently, with no throttle or node constraint. Resume by validating both outcomes and cleanup, then isolate the `binary_add` records between Sweep begin/end and compare length, ordered/set hashes, the first mismatch, and both target addition ordinals.

Both tasks completed as valid UNKNOWN resource timeouts with untruncated traces, deleted solver artifacts, and no pending cleanup. Kissat's run is 120.117 seconds with a 90,112-byte trace (SHA-256 prefix `d8dbc884`); Tactix's is 120.468 seconds with a 21,173-byte trace (prefix `8b0c5202`). Between Sweep begin/end, both solvers publish exactly the same four direct additions, in the same order: `(-28,3825745)`, `(-28,-39)`, `(-28,-48)`, and `(-28,5913)`. Thus the six-to-264/260 watch-list

growth is not a new-binary insertion mismatch: Sweep equivalence substitution moves or republishes pre-existing binary occurrences onto representative -28. Resume by tracing that representative's retiring-source classes and comparing source emission order, each source stream's encounter order, and the four members present only in Kissat's Sweep-end list.

The selected-representative stream observer is implemented in commit afd154d. Under SAT_EVENT_TRACE_SWEEP_ each solver now records only binary occurrences actually appended to representative -28, naming the retiring source literal, its unified-stream position, and the resulting partner. Search-only mode retains these records; the option is otherwise disabled and production compilation erases the Tactix calls. The complete Trace suite passes 71/71, the isolated Release suite passes 72/72, static actual-hook assertions pass, and a fresh optimized Kissat overlay builds cleanly. The Tactix/Kissat observer SHA-256 prefixes are 5b1857eb/b0591cea. Resume by pinning them in a two-arm Slurm campaign and comparing moved-stream counts, exact partner order, source-block order, first mismatch, and the four Kissat-only members.

The paired stream capture is campaign

20260819-1016-tierd-rank22-sweep-streams

and Slurm job 1634583, tasks 0–1; scheduler dry run 1634582 passed. Each independent task has two CPUs, 16 GiB, and a 120-second solver ceiling, with no throttle or node constraint. The immutable Tactix/Kissat pins have SHA-256 prefixes 5b1857eb/b0591cea. Resume by waiting for job 1634583, requiring valid outcomes, untruncated traces, and complete cleanup, then comparing moved counts, exact partner order, retiring-source blocks, first mismatch, both target ordinals, and the Kissat-only members.

Both tasks completed as valid UNKNOWN resource timeouts with untruncated traces and complete cleanup. Kissat ran 120.118 seconds and wrote 131,072 bytes (SHA-256 prefix 69913119); Tactix ran 120.468 seconds and wrote 59,207 bytes (prefix 4a6ef83b). Kissat records 254 moved binaries and Tactix 255. Both use the same source-block order, -3825745, 39, then -5913, with counts 12/240/2 versus 13/239/3. Both targets occur at the same global move ordinals, 4 and 243. The first mismatch is ordinal 12: Kissat moves -3825772 from source position 25, while Tactix moves -3859674 from position 11. The complete multiset delta is small: Kissat alone has -132 and -3825772; Tactix alone has -30, -3859674, and -3859764. Partner-sequence SHA-256 prefixes are c4607a15/f75e2c65. Resume by inspecting the physical binary-watch rewrite against the already source-ordered Sweep ledger and explaining why Tactix's 255 logged moves produce only 250 net new physical -28 entries before changing behavior.

The resulting repair is commit 1e0fa6d. Tactix already computed the reference emission order and used it for the logical dense Sweep ledger, but the physical binary-watch reconstruction independently walked retiring literals in global import order. That left later Sweep environments correct through the ledger while sparse search consumed a different physical order. The connected Sweep rewrite now snapshots directed binary images in the session's retiring-source order; canonical Substitute retains its existing import-order traversal. The former ledger-only emission-order regression now also requires the physical representative list to contain the same source blocks. Trace passes 71/71 and isolated Release passes 72/72. Resume with a focused Slurm verification requiring the Sweep-end -28 target pose to move toward Kissat and the conflict-611 assignment/trail order to converge; retain the repair only if both guards stay clean.

The focused verification is campaign

20260819-1026-tierd-rank22-emission-order-verify

and Slurm job 1634586; scheduler dry run 1634585 passed. One immutable Tactix observer pin (SHA-256 prefix 01f62a79) runs with two CPUs, 16 GiB, a 120-second solver ceiling, no throttle,

and no node constraint. It retains both the Sweep-end compact watch summary and conflict-611 detail, then stops immediately after the selected analysis. Resume by requiring a valid UNKNOWN stop, an untruncated trace, complete cleanup, and both summaries; compare them with saved Kissat jobs `1634577` and `1634574` respectively.

Job `1634586` completes by the selected stop in 82.55 seconds as a valid UNKNOWN with a 27,104-byte untruncated trace (SHA-256 prefix `b1e86d5c`), deleted artifact, and no pending cleanup. At conflict 611, the repaired Tactix list still has 261 binaries but `-3858960` moves from position 248 to 9, exactly Kissat's position; `-3858955` moves from 235 to 248, versus Kissat 246. The formerly reversed target order is therefore repaired. Decision 11409, selected external variable 3730173, and the complete stable top-32 order match saved Kissat, although the full watch hash and wider state digest remain unequal. Endpoint-only stop suppression deliberately omits the pre-target Sweep-end snapshot; the stronger conflict-611 endpoint is present. Retain the repair for a proof-producing production rank-22 arm under the iteration resource contract, then require the paired retained-rank guard before acceptance.

The production validation is campaign

`20260819-1031-tierd-rank22-production`

and Slurm job `1634588`; scheduler dry run `1634587` passed. One immutable optimized Tactix pin (SHA-256 prefix `d0069191`) runs with two CPUs, 16 GiB, a 1,200-second solver ceiling, no throttle, and no node constraint. The frozen reference result is UNSAT in 196.556 seconds. Acceptance requires Tactix to report UNSAT, independent `drat-trim` verification, measured wall/RSS, deleted proof/input scratch, and no pending cleanup. Resume by waiting for job `1634588`; if it passes, run the paired retained-rank guard before accepting `1e0fa6d`.

Job `1634588` is a valid clean timeout, not a recovered solve. The optimized candidate reaches the 1,200-second ceiling at 1,200.004 seconds and reports UNKNOWN. Its incomplete 108,003,328-byte proof artifact (SHA-256 prefix `4e846c61`) is not checked or accepted; the decompressed input, partial proof, and scratch are deleted, with no pending cleanup. Thus the emission-order repair corrects the known conflict-611 target ordering but is insufficient to close frozen rank 22 under the iteration contract. Do not run the retained-rank guard or accept the repair yet. Keep rank 22 open and resume paired semantic divergence tracing immediately after the now-aligned conflict-611 target order.

That paired trace is prepared as campaign

`20260819-1056-tierd-rank22-post611-trajectory`

It freezes the repaired Tactix observer (SHA-256 prefix `01f62a79`) and the unchanged Kissat trajectory observer (prefix `4fe71d17`) over conflicts 612–1,200. Both independent tasks request two CPUs, 16 GiB, a 300-second solver ceiling, no throttle, and no node constraint. Trajectory-only mode retains decisions, conflicts, analysis boundaries, random/phase choices, and constant-size learned-clause summaries while suppressing watch and implication volume. Resume by dry-validating and submitting the array, then require valid outcomes, untruncated traces, complete cleanup, and the first normalized portable mismatch. The prior frontier was exact through conflict 1,198; this campaign determines how the conflict-611 physical-order repair changes that frontier. Scheduler dry validation passed as job `1634591`; the two-task campaign is now Slurm job `1634592`. Resume by waiting for both tasks, validating the outcomes, trace completeness, and cleanup, then locating the first normalized portable mismatch.

Both job `1634592` tasks are valid, clean UNKNOWN timeouts. Kissat runs 300.125 seconds and publishes a 5,177,344-byte trace (SHA-256 prefix `8908bf1c`); Tactix runs 300.380 seconds and publishes a terminal 8,235,567-byte trace (prefix `581ea521`). The complete requested windows

reach conflict 1,200 without a truncation marker, and both tasks delete all scratch with no pending cleanup. After ignoring the known compact `analysis_end` snapshot asymmetry, each window has 18,322 normalized decision, conflict, analysis-boundary, and learned-summary records. They match through conflict 1,199 analysis and its first decision `-3779280`. The next decision is the unchanged frontier: at decision 25,781 and level 469, Kissat selects `-3782012` while Tactix selects `-3782027`. Thus the conflict-611 physical-order edit fixes its targeted list ordering, but search reconverges by conflict 612 and the later causal frontier does not advance. Keep `1e0fa6d` unaccepted and rank 22 open. Resume by comparing the full repaired conflict-611 `-28` list with Kissat, beginning with the residual two-position offset of `-3858955`, and test whether it preserves the known conflict-1,001 reinsertion/heap mismatch.

Comparing the saved Sweep rewrite streams reduces the residual further. After dropping set-only entries, all 252 common binary partners occur in exactly the same relative order, including both target partners. Kissat alone moves `-3825772` from source `-3825745` and `-132` from source `39`; Tactix alone moves `-3859674` and `-3859764` from `-3825745`, plus `-30` from `-5913`. The latter three pairs are literal input binaries, so representative-map filtering rather than another common-partner ordering error is now the leading discriminator. Search-only tracing now retains the already-existing `sweep_equivalence` event symmetrically; the Kissat materializer inserts its corresponding hook. Focused/static tests pass, followed by Trace 71/71 and Release 72/72, and a fresh Kissat overlay builds. Observer SHA-256 prefixes are `e06a764b` (Tactix) and `bfea3d25` (Kissat). Resume with a paired conflict-zero search-only trace that records equivalence pairs alongside the existing `-28` rewrite stream, then map the five set-only partners to their class representatives before considering any solver edit.

The immutable paired campaign is prepared at

`20260819-1113-tierd-rank22-sweep-equivalences`

It combines both equivalence streams with the existing selected `sweep_rewrite_binary` stream at conflict zero. Each task requests two CPUs, 16 GiB, a 120-second solver ceiling, no throttle, and no node constraint. Resume by scheduler dry validation and submission, then require valid outcomes, untruncated traces, deleted scratch, and representative chains for all five set-only partners. Scheduler dry validation passed as job `1634594`; the recorded two-task campaign is now Slurm job `1634595`. Resume by waiting for both arms, validating trace completeness and cleanup, then mapping all five partners.

Both job `1634595` tasks are valid, residue-free UNKNOWN timeouts with untruncated traces and deleted scratch. Kissat runs 120.118 seconds and emits 248 equivalence plus 254 selected rewrite records in 163,840 bytes (SHA-256 prefix `bd41bf23`); Tactix runs 120.418 seconds and emits 229 plus 255 records in a terminal 86,999-byte trace (prefix `64152642`). After normalizing endpoint orientation and mirror polarity, every Tactix equivalence appears in Kissat; Kissat alone emits 19. The first missing cluster is `(3825774,-3825776)`, `(-3820683,3825774)`, and `(3826332,-3820683)`. A later five-pair class includes `(-30,-3809178)`, explaining why Kissat retires input binary `(-5913,-30)` before `-5913` merges into `-28`, while Tactix moves it. Both streams do contain the common classes connecting `3825772` and `132` to the eventual `-28` class. Thus Tactix's equivalence set is a strict subset and clause-retirement timing explains the membership residual; common-partner order is exonerated. This is simplification-performance evidence, not a soundness finding. Resume by adding a symmetric search-only seed record, identify the reference seed owning the first missing triple, and compare only that seed's environment/candidate path.

Search-only tracing now also retains `sweep_seed`. Tactix emits the existing seed hook; the Kissat materializer inserts the symmetric event at `sweep_variable` entry before environment construction. Focused and static coverage pass, followed by Trace 71/71, Release 72/72, and a fresh Kissat build.

Observer SHA-256 prefixes are `9437950a` (Tactix) and `f865ce98` (Kissat). Resume with one paired conflict-zero seed/equivalence trace and attach each extra Kissat pair to the immediately preceding seed.

That paired capture is campaign

`20260819-1123-tierd-rank22-sweep-seeds`

with two CPUs and 16 GiB per independent task, a 120-second ceiling, no throttle, and no node constraint. Scheduler dry validation passed as job `1634600`. Resume by submitting the pair, then validate and bind each equivalence to its preceding seed. The pair is now Slurm job `1634601`; resume by waiting for both tasks and performing the recorded validation and binding.

Both job `1634601` arms are valid, untruncated, residue-free UNKNOWN timeouts with complete cleanup. Kissat runs 120.118 seconds and publishes a 188,416-byte trace (SHA-256 prefix `8219b1c2`); Tactix runs 120.418 seconds and publishes a terminal 114,017-byte trace (prefix `b1473f8c`). Both visit exactly 243 external seeds in identical order. Only seed ordinals 51–57 and 77 have different equivalence groups, always Kissat-only additions. At the first differing group, seed `3826352`, both emit `(-3826352,-3826351)` and `(-3859775,-3859774)`; Kissat alone then emits `(-3825776,3825774)`, `(-3825774,3820683)`, and `(-3826332,3820683)`. Seed scheduling and prior representative rescheduling are therefore exonerated. Resume with a symmetric search-only environment-size record for seed `3826352`; if counts match, compare its initial candidate partition and selected queries.

The symmetric search-only `sweep_environment` observer is now implemented without changing solver semantics. Tactix retains its existing environment event; the Kissat materializer emits external seed, variable count, and encoded clause count immediately after environment construction. Focused/static tests, Trace 71/71, Release 72/72, and a fresh Kissat observer build all pass. SHA-256 prefixes are `5bad7b9b` (Tactix) and `5edbacca` (Kissat). Resume with one paired conflict-zero capture and compare the environment counts at external seed `3826352`; only if they match should the next observer expose content or candidate/model/query selection.

The predeclared paired capture is campaign

`20260819-1133-tierd-rank22-sweep-environments`

with immutable observer pins, two CPUs and 16 GiB per independent task, a 120-second solver ceiling, no throttle, and no node constraint. Scheduler dry validation passed as job `1634607`. Resume by submitting the pair, then validate both results and compare only external seed, environment variable count, and encoded-clause count at seed `3826352`. The submitted pair is Slurm job `1634608`. Resume by waiting for both array tasks, performing the recorded validity/cleanup checks, and making that three-field comparison before adding any broader content observer.

Both job `1634608` arms completed as valid, residue-free UNKNOWN timeouts with untruncated traces and complete scratch cleanup. Kissat ran 120.117 seconds and emitted 229,376 bytes (SHA-256 prefix `35d2ad39`); Tactix ran 120.418 seconds and emitted a terminal 151,892 bytes (prefix `73ca2099`). All 243 ordered triples of external seed, environment-variable count, and encoded-clause count match exactly. At external seed `3826352`, each environment contains 26 variables and 38 clauses. Environment cardinality and construction schedule are therefore exonerated. Resume with a bounded symmetric comparison of that seed's environment content/order, then initial model/candidate/query selection.

The next observer reuses Tactix's existing seed-filtered `sweep_environment_clause` hook and adds the symmetric Kissat record at the exact handoff of each selected clause to Kitten. It records clause ordinal and external literal order only for the seed selected by `SAT_EVENT_TRACE_SWEEP_CLAUSES`; search-only mode now retains the event. Focused/static coverage, Trace 71/71, Release 72/72, and a fresh Kissat build pass. Observer SHA-256 prefixes are `dc1c33a9` (Tactix) and `5fb5409d` (Kissat).

This is bounded diagnostic instrumentation, not a solver-semantic or soundness change. Resume with one paired seed-3826352 clause capture and compare ordered clauses first, then clause sets if order differs.

That paired capture is campaign

`20260819-1143-tierd-rank22-sweep-clauses`

with the common seed filter set to `3826352`, two CPUs and 16 GiB per independent task, a 120-second ceiling, no throttle, and no node constraint. Scheduler dry validation passed as job `1634613`. Resume by submitting the pair, validating both arms, and comparing the 38 ordered environment clauses; if order differs, separately compare their normalized literal sets. The submitted pair is Slurm job `1634614`. Resume by waiting for both tasks, applying the recorded validity and cleanup checks, and performing those ordered and set-normalized clause comparisons.

Both job `1634614` arms are valid, residue-free UNKNOWN timeouts with untruncated traces and complete scratch cleanup. Kissat runs 120.118 seconds and emits 229,376 bytes (SHA-256 prefix `617be092`); Tactix runs 120.519 seconds and emits a terminal 163,076 bytes (prefix `56d1191d`). Each arm records 38 environment clauses for seed `3826352`. Every clause ordinal, every external literal, and every within-clause position matches exactly; the normalized clause sequence and multiset are therefore exact as well. Environment membership and both levels of encoding order are exonerated. Resume with a symmetric seed-filtered initial Kitten result/model record, then compare the candidate-query sequence only if that initial state matches.

Search-only tracing now retains Tactix's existing seed-filtered `sweep_query` and `sweep_query_model` events. The Kissat overlay adds the symmetric records around every Sweep subsolver call: initial, backbone, forward equivalence, and reverse equivalence. Kissat results are normalized from `10/20/0` to Tactix's `Sat/Unsat/Budget=0/1/2`; candidate and model literals are externalized, and model order follows the environment variable stream. Focused/static coverage, Trace 71/71, Release 72/72, and a fresh Kissat build pass. Observer SHA-256 prefixes are `b19a3f4b` (Tactix) and `b4e0d93d` (Kissat). Resume with one paired seed-3826352 query/model capture and locate the first mismatch while ignoring implementation-specific tick totals.

That paired capture is campaign

`20260819-1153-tierd-rank22-sweep-queries`

with common clause and query filters set to `3826352`, two CPUs and 16 GiB per independent task, a 120-second ceiling, no throttle, and no node constraint. Scheduler dry validation passed as job `1634616`. Resume by submitting the pair, validating both arms, and comparing ordinal, kind, candidate literals, normalized result, and model while ignoring ticks and implementation-specific conflict counters. The submitted pair is Slurm job `1634617`. Resume by waiting for both tasks, performing the recorded validity/cleanup checks, and locating the first query or model mismatch under that normalization.

Both job `1634617` arms are valid, residue-free UNKNOWN timeouts with untruncated traces and complete cleanup. Kissat runs 120.118 seconds and emits 237,568 bytes (SHA-256 prefix `0feb9b16`); Tactix runs 120.419 seconds and emits a terminal 163,971 bytes (prefix `c46a6d47`). All 22 query records match exactly in ordinal, kind, candidate literals, and normalized result, and all eight SAT models match exactly as ordered external signed-literal streams. At ordinals 16–21 both solvers refute both directions of `(3825774,-3825776)`, `(-3820683,3825774)`, and `(3826332,-3820683)`, yet only Kissat commits those equivalences. The first divergence is therefore after semantic refutation, in Tactix's captured-proof selection/translation or consequence-commit path. Resume with a conflict-zero Sweep rule-journal capture filtered to candidate `3825774,-3825776` and inspect its structured decline.

The predeclared one-arm diagnostic is campaign

`20260819-1159-tierd-rank22-sweep-commit-journal`

using the validated immutable Tactix trace pin, two CPUs, 16 GiB, a 120-second solver ceiling, no throttle, and no node constraint. The existing rule journal is restricted to rule `Sweep`, conflict zero, candidate `3825774,-3825776`, and 20 records. Scheduler dry validation passed as job `1634619`. Resume by submitting the task, validating cleanup and journal integrity, then reading its consequence-select/commit decline fields. The submitted diagnostic is Slurm job `1634620`. Resume by waiting for the task and carrying out exactly those journal-integrity and decline checks.

Job `1634620` completes as a valid, residue-free UNKNOWN timeout in 121.370 seconds with a terminal untruncated 153,020-byte event trace (SHA-256 prefix `bf077246`), but creates no journal; the result metadata correctly records null journal bytes, hash, and record count. Inspection shows that the pinned `build/trace` binary enables event tracing but has `TACTIX_ENABLE_RULE_JOURNAL=OFF`; only the standalone journal test defines it. This is an observer-configuration miss and supports no solver conclusion. Resume by building and validating a combined event-trace plus rule-journal pin, then rerun the identical candidate-filtered task.

A separate `build/trace-journal` is now configured with both `TACTIX_ENABLE_EVENT_TRACE=ON` and `TACTIX_ENABLE_RULE_JOURNAL=ON`. Its complete 72-test suite passes, including the journal-enabled default-policy integration test. The corrected binary's SHA-256 prefix is `b3cf12b4`. Resume by pinning this binary into a fresh immutable campaign and rerunning the identical conflict-zero candidate filter; do not reinterpret job `1634620`.

The corrected campaign is

`20260819-1207-tierd-rank22-sweep-commit-journal`

and its manifest explicitly records both enabled compile flags and pin hash. It keeps the same candidate, conflict-zero, 20-record, 2-CPU, 16 GiB, and 120-second contracts, with no throttle or node constraint. Scheduler dry validation passed as job `1634621`. Resume by submitting this corrected task, then validate and interpret its journal artifact. The submitted corrected diagnostic is Slurm job `1634622`. Resume by waiting for the task, validating journal bytes/hash/record count and cleanup, then reading the candidate's structured selection and commit disposition.

Job `1634622` completes as a valid, residue-free UNKNOWN timeout in 121.070 seconds with a terminal untruncated 153,020-byte event trace (SHA-256 prefix `bf077246`). The corrected binary creates a valid 203-byte journal (prefix `42ceb79f`), but it contains only the two headers and zero data records. Thus the assumed candidate spelling/order did not match the stored journal key, or this path emitted no candidate record; no commitment conclusion follows. Resume with the same conflict-zero `Sweep` journal without a candidate filter, bounded to 200 records, then identify the target spelling and structured decline from that finite stream.

The replacement campaign is

`20260819-1213-tierd-rank22-sweep-commit-journal-all`

using the same combined pin and resource contract. It retains rule `Sweep` and conflict zero but removes the candidate filter, with a hard 200-record bound. Scheduler dry validation passed as job `1634623`. Resume by submitting, validating, then finding records whose candidate literal set is `{3825774,-3825776}` independent of stored order. The submitted bounded task is Slurm job `1634624`. Resume by waiting for it, validating journal integrity and cleanup, then carrying out that order-independent candidate lookup.

Job `1634624` completes as a valid, residue-free UNKNOWN timeout in 120.769 seconds with the same terminal untruncated 153,020-byte trace and a valid 203-byte journal, but the unfiltered,

200-record-bounded journal still has zero data records. Therefore no conflict-zero Sweep consequence reaches the journaled selection/general-plan path in this run. The target exits earlier: fixed/no-proof unit collapse, degenerate/known handling, root-state gating, or a successful direct commit whose event semantics omit that branch. Resume by adding a pair-filtered Tactix `sweep_commit` event at entry and each pre-journal branch for unordered pair `{3825774,-3825776}`, recording both endpoint root values and the selected outcome.

That bounded observer is now implemented behind `SAT_EVENT_TRACE_SWEEP_COMMIT`. It accepts the unordered external pair, records endpoint values, proof mode, and entry/early-exit/commit outcome, and remains available in search-only traces; without the environment filter it emits nothing. The focused test and the full 72/72 release plus 71/71 trace regressions pass. Resume by committing the observer and this checkpoint, then preparing, dry-validating, and submitting one 16 GiB rank-22 branch trace.

The observer is committed as `893dda7`; its immutable trace binary has SHA-256 prefix `8e14dfdb`. Campaign

`20260819-1225-tierd-rank22-sweep-commit-branch`

pins that binary and the checked rank-22 input, requests 16 GiB and two CPUs with a 120-second solver ceiling, no throttle, and no node constraint. Scheduler dry validation passed as job `1634625`. Resume by submitting the single task, recording its job ID, and waiting for the terminal validated result.

The actual one-task diagnostic is Slurm job `1634626`. Resume by waiting for it, then require a valid checked answer or clean UNKNOWN timeout, a terminal untruncated trace, at least one pair-filtered `sweep_commit` record, and confirmed scratch cleanup before interpreting the branch.

Job `1634626` completes as a valid, residue-free UNKNOWN timeout in 120.519 seconds with a terminal untruncated 117,447-byte trace (SHA-256 prefix `d7ade4d8`). The target produces exactly two branch records: entry with `av=1`, `bv=1`, `proof=0`, followed by `fixed-no-proof-no-unit`. Both endpoints are already root-true. Tactix does retain this compatible fixed pair in the emitted-equivalence substitution batch; only its early branch omits the `sweep_equivalence` event. Thus the apparent first missing consequence is observer-semantic asymmetry, not a discarded solver consequence. Resume by emitting that semantic event whenever the branch appends the pair, validating both suites, and rerunning the bounded source-order comparison to find the next genuine mismatch.

The observer correction now emits `sweep_equivalence` exactly when the fixed/no-proof branch appends the compatible pair to the substitution batch; incompatible fixed pairs remain unreported and uncommitted. Focused Sweep and event-trace tests pass. Resume by completing the full release and trace suites, then commit and rerun the bounded rank-22 source-order comparison.

The complete validation is green: 72/72 release tests and 71/71 event-trace tests. Resume by committing this observer-only semantic correction, pinning the fresh trace binary, and rerunning the rank-22 conflict-zero source-order stream against the saved immutable Kissat task from job `1634595`.

The correction is committed as `470b33f`; fresh Tactix trace binary SHA-256 prefix `1734a826`. Campaign

`20260819-1233-tierd-rank22-sweep-equivalences-corrected`

uses one 16 GiB, two-CPU, 120-second unthrottled task and compares it with the saved Kissat trace SHA-256 prefix `bd41bf23`. Scheduler dry validation passed as job `1634627`. Resume by submitting and waiting for the corrected Tactix task, then normalize and compare both equivalence streams.

The actual corrected-stream task is Slurm job `1634628`. Resume by waiting for it, validating answer/trace/cleanup integrity, and comparing its normalized `sweep_equivalence` sequence with the saved Kissat stream.

Job `1634628` completes as a valid, residue-free UNKNOWN timeout in 120.419 seconds with a terminal untruncated 120,872-byte trace (SHA-256 prefix `cf6e915c`). Corrected Tactix and the immutable Kissat reference each emit 248 normalized Sweep equivalences. The normalized sets and the complete normalized sequence are exact, with no duplicates and no one-sided pairs. Therefore the former 19-pair strict-subset result was entirely observer-semantic asymmetry. Detection and retained-equivalence source order are exonerated; the remaining rank-22 divergence is downstream in substitution/rewrite application or later search. Resume at the exact equivalence boundary by comparing the selected `-28` rewrite stream under the corrected source-order build.

The saved job-`1634595` selected-rewrite streams remain valid because the observer correction changes only equivalence reporting. Ignoring position shifts, Kissat alone moves `(-3825745,-28,-3825772)` and `(39,-28,-132)`; Tactix alone moves the two high-variable partners from source `-3825745` plus `(-5913,-28,-30)`. The now-exact pair order and source code identify the architectural boundary: Kissat substitutes each proved pair immediately, while Tactix applies one final-class batch and then reconstructs source order. That residual does not change the portable search trajectory before the known post-analysis conflict-1199 decision frontier. Resume with symmetric endpoint-stop traces at conflict 1199, capturing the full stable heap immediately after analysis and before divergent decision 25,781.

Campaign

`20260819-1243-tierd-rank22-c1199-heap-endpoint`

pins the validated Kissat trajectory observer (SHA-256 prefix `4fe71d17`) and corrected Tactix observer (prefix `1734a826`). Its two independent, unthrottled tasks each request 16 GiB, two CPUs, and a 300-second ceiling, then stop immediately after conflict-1199 analysis. Scheduler dry validation passed as job `1634630`. Resume by submitting, waiting, and comparing the stable heap hashes, top order, and scores of externals 3782012 and 3782027.

The actual symmetric endpoint array is Slurm job `1634631`. Resume by waiting for both tasks and applying the recorded endpoint integrity checks before interpreting either heap.

Job `1634631` is a partial diagnostic result. Tactix stops correctly at conflict 1199 in 97.863 seconds with a terminal untruncated 218,712-byte trace (SHA-256 prefix `ee836a11`), stable heap hash `2dcd85762dac1a9f`, and complete cleanup. The selected Kissat pin crosses the target but its endpoint control fails to terminate: it emits repeated heaps, continues to conflict 65,820, and hits the Slurm limit without a structured result. This is an observer-control miss, not solver evidence. Preserve the valid Tactix endpoint and retry only Kissat with the already validated endpoint pin SHA-256 prefix `682a4e7c` from job `1634505`; then compare the single target heap with the saved Tactix result.

The one-task retry campaign is

`20260819-1255-tierd-rank22-kissat-c1199-heap-retry`

using validated endpoint pin `682a4e7c`, 16 GiB, two CPUs, a 300-second solver ceiling, no throttle, and no node constraint. Scheduler dry validation passed as job `1634635`. Resume by submitting and waiting for this Kissat task only, then compare it with immutable Tactix trace `ee836a11`.

The actual one-task retry is Slurm job `1634636`. Resume by waiting for it and applying the endpoint integrity checks before the saved-trace comparison.

Job `1634636` succeeds in 29.132 seconds as a valid intentional-stop UNKNOWN with an untruncated 222,387-byte trace (SHA-256 prefix `cee2bb3a`) and complete cleanup. At conflict 1199

both endpoints have decision count 25,779, level 467, and exactly the same normalized top-32 candidate order; external 3782012 immediately precedes 3782027. Their raw heap hashes differ: Kissat `259d1d77227e9979`, Tactix `2dcd85762dac1a9f`. Both next choose `-3779280` at decision 25,780, then split. The earliest live discriminator is therefore the lazy heap removal/sift after that common decision, not analysis-end score ranking. Resume with heap-pop-only traces for Kissat decision 25,781 and Tactix's corresponding pre-increment decision 25,780.

Campaign

`20260819-1259-tierd-rank22-d25781-heap-pop`

pins the two current observers and retains only the targeted lazy-pop path at conflict 1199: Kissat decision 25,781 and Tactix pre-increment decision 25,780. Both tasks use 16 GiB, two CPUs, a 120-second ceiling, no throttle, and no node constraint. Scheduler dry validation passed as job `1634637`. Resume by submitting, waiting, then comparing begin/move/end identities, scores, and heap indices.

The actual two-task heap-pop array is Slurm job `1634638`. Resume by waiting for both tasks and validating trace/cleanup integrity before comparing the filtered pop paths.

Job `1634638` produces two valid, residue-free UNKNOWN timeouts and untruncated traces, but zero `heap_pop` records on both sides. The target was late. Immediately after conflict-1199 analysis, restart-reuse policy asks for the next unassigned heap variable while the decision counter is still 25,779; that call lazily removes the assigned maximum before either next decision is counted. This is a targeting miss, not solver evidence. Resume by repeating the heap-pop-only comparison with counter 25,779 for both solvers.

The corrected campaign is

`20260819-1305-tierd-rank22-restart-reuse-heap-pop`

with the same immutable pins and resources but counter 25,779 on both tasks. Scheduler dry validation passed as job `1634640`. Resume by submitting, waiting, and comparing the restart-reuse pop paths.

The actual corrected array is Slurm job `1634641`. Resume by waiting for both tasks and validating the two filtered paths plus cleanup.

Job `1634641` is also clean and empty. The counter was not tested: `heap_pop_only` passes the event through its first filter, but `search_only` then drops `heap_pop` because it is not whitelisted. The two target attempts have identical per-solver trace hashes, confirming a configuration miss. Resume with the original Kissat-25,781/Tactix-25,780 targets, heap-pop-only enabled, and search-only disabled.

The corrected visible-event campaign is

`20260819-1311-tierd-rank22-d25781-heap-pop-visible`

using the original decision targets, heap-pop-only filtering, and no search-only filter. Scheduler dry validation passed as job `1634643`; the actual array is job `1634644`. Resume by waiting and comparing the retained paths.

Job `1634644` completed as two clean, bounded UNKNOWN timeouts with independent scratch cleanup. Kissat ran 120.118 seconds and wrote an untruncated 704,512-byte trace (SHA-256 prefix `4bb0ab8d`) but retained no `heap_pop` records: immutable pin `4fe71d17` predates that hook. Tactix ran 120.318 seconds and wrote an untruncated 833,558-byte trace (SHA-256 prefix `d933d462`) containing 28 consecutive lazy pops before the split. Its final pop promotes external variable 3,782,027 ahead of 3,782,012 even though both scores are exactly 25774.157807429845. Preserve that

valid Tactix trace and resume by rerunning only Kissat with heap-pop-capable pin `8ce35871` from the 08:57 campaign, then compare the first differing pop path.

That isolated continuation is campaign

`20260819-1318-tierd-rank22-kissat-d25781-heap-pop-retry`

with the heap-pop-capable Kissat pin, one task, the original decision-25,781 target, heap-pop-only filtering, and no search-only filter. Scheduler dry validation passed as job `1634646`. Resume by submitting this campaign, waiting for its bounded result, and comparing its begin/move/end path with the saved Tactix trace whose SHA-256 prefix is `d933d462`.

The actual isolated retry is Slurm job `1634647`. Resume by waiting for that task, validating its bounded answer, trace integrity, and scratch cleanup, then comparing its retained pop path with the saved Tactix trace.

Job `1634647` is a valid clean bounded UNKNOWN: 120.119 seconds, an untruncated 851,968-byte trace (SHA-256 prefix `12194e9b`), deleted artifact, and no pending cleanup. Kissat and the saved Tactix arm each lazily pop 28 variables, and the popped-variable sets are identical. Their first pop paths agree through seven child moves; the next move is Kissat variable 3,778,844 from index 410 to 204 versus Tactix variable 3,778,692 from index 408 to 203. Pop-root order first differs at operation 12. On the final relevant promotion, Kissat moves 3,782,012 above equal-score 3,782,027 while Tactix moves 3,782,027 above 3,782,012; both scores are 25774.157807429845. This exonerates the heap comparator and score values: the decision split is caused by the raw heap pose already known to differ. Resume by comparing the repaired Tactix and Kissat conflict-1,001 heap-reinsertion streams to locate the earlier ordering mutation that creates this pose.

That comparison is campaign

`20260819-1325-tierd-rank22-repaired-reinsert-sequence`

using immutable Kissat/Tactix observer pins `8ce35871`/`1734a826`, two CPUs and 16 GiB independently per task, a 120-second solver ceiling, no throttle, and no node constraint. Scheduler dry validation passed as job `1634648`; the actual pair is Slurm job `1634649`. Resume by waiting for both arms, validating answer/trace/cleanup integrity, then comparing conflict-1,001 `heap_reinsert` counts, membership, ordered hashes, and the first mismatch against the pre-repair item-43 split.

Both job `1634649` tasks are valid, untruncated, and residue-free UNKNOWN outcomes. Kissat times out cleanly in 120.136 seconds with 11,034,624 trace bytes (SHA-256 prefix `8dee55ad`); Tactix stops terminally in 96.980 seconds with 10,260,808 bytes (prefix `8528cc81`). Each conflict-1,001 reinsertion stream has 32,690 external variables. Their membership and exact order match, with ordered SHA-256 prefix `50a4ab54`; variables 3,858,960 and 3,858,955 are at positions 43 and 48 in both. Before the repair, Tactix put them at positions 51 and 43 and first differed at item 43. Thus the repair fully fixes this causal checkpoint, but the raw heap later diverges again. Resume by binary-searching raw heap endpoint hashes between the exact post-conflict-1,001 state and the known conflict-1,199 mismatch.

The coarse endpoint array is campaign

`20260819-1331-tierd-rank22-heap-endpoint-coarse`

at paired conflicts 1,002, 1,050, 1,100, 1,150, and 1,198. All ten immutable tasks run independently with two CPUs and 16 GiB, a 120-second solver ceiling, no throttle, and no node constraint. Scheduler dry validation passed as job `1634652`; the actual array is Slurm job `1634653`. Resume by waiting for every task, validating answer/trace/cleanup integrity, and bracketing the first unequal raw heap hash while requiring normalized top-32 parity.

All ten job 1634653 tasks have valid UNKNOWN answers, untruncated traces, deleted artifacts, and no pending cleanup. The conflict-1,002 Kissat arm reaches 120.117 seconds before publishing its endpoint because observer pin 8ce35871 snapshots that large selected analysis; Tactix's stop lands at conflict 1,003 instead, so the earliest pair is inconclusive. Every later sample already differs: the Kissat/Tactix raw hashes at conflicts 1,050, 1,100, 1,150, and 1,198 are respectively a5865471/a388013c, c9f3cf06/ab50a012, 4baf98c4/750314cd, and c9daee7b/70a2a5ee. Their top-32 poses also differ, although the known conflict-1,199 top-32 later reconverges while its raw pose remains different. Resume by refining conflicts 1,003–1,050 with lighter validated Kissat endpoint pin 682a4e7c and the same repaired Tactix pin.

The refinement campaign is

20260819-1336-tierd-rank22-heap-endpoint-refine

It runs the lighter Kissat pin at conflict 1,003 against the saved Tactix endpoint, then pairs conflicts 1,010, 1,020, 1,030, and 1,040. Nine immutable tasks retain the same independent two-CPU/16-GiB contract, with no throttle or node constraint. Scheduler dry validation passed as job 1634664; the actual array is Slurm job 1634665. Resume by waiting for every task, validating answer/trace/cleanup integrity, and bracketing the first unequal endpoint between conflicts 1,003 and 1,050.

All nine job 1634665 tasks have valid UNKNOWN answers, untruncated traces, deleted artifacts, and no pending cleanup. Kissat's conflict-1,003 raw heap hash 2cc932e10ca5ec55 exactly matches the saved Tactix endpoint. By conflict 1,010 the hashes differ, c69bd8a2baab90fe versus 79dc1ce4f6f0fae4, and top-32 order first differs at position 2. Conflicts 1,020 and 1,040 also differ. The conflict-1,030 pair publishes no exact-target endpoint because the selected analysis/stop semantics overshoot, but it lies outside the proven bracket. Resume by sampling every conflict from 1,004 through 1,009 with the same lighter endpoint pins.

The exact-conflict campaign is

20260819-1341-tierd-rank22-first-heap-mismatch

and pairs every endpoint from conflict 1,004 through 1,009. Twelve immutable tasks use the same independent two-CPU/16-GiB contract, with no throttle or node constraint. Scheduler dry validation passed as job 1634674; the actual array is Slurm job 1634675. Resume by waiting for all tasks, validating answer/trace/cleanup integrity, and selecting the first unequal paired endpoint as the causal conflict.

All twelve job 1634675 tasks are valid intentional-stop UNKNOWNs with untruncated traces, deleted artifacts, and no pending cleanup. Raw heap hashes match exactly at conflicts 1,004, 1,005, and 1,006: c347670f0d35b6c8, f630e2a89e515c73, and cb7647e239c08a03. The first mismatch is conflict 1,007, where Kissat has 2dc7247556f5f30b and Tactix has d059788fa9995d69; conflicts 1,008 and 1,009 remain unequal. The complete conflict-1,007 top-32 identity order is still exact. Resume inside conflict 1,007 by comparing the pre-bump heap pose, ordered score-bump identities, and any backtrack reinsertion stream; the first unequal internal checkpoint is the next repair target.

The per-bump comparison is campaign

20260819-1347-tierd-rank22-c1007-bump-heaps

It enables the existing diagnostic-only heap checkpoint before bumping and after each common variable 749,485, 743,678, 742,665, and 746,437. Both immutable tasks retain the independent two-CPU/16-GiB contract, no throttle, and no node constraint. Scheduler dry validation passed as job 1634687; the actual pair is Slurm job 1634688. Resume by waiting for both arms, validating answer/trace/cleanup integrity, then aligning the pre-bump and four post-bump raw hashes.

Both job 1634688 tasks are valid, untruncated, and residue-free UNKNOWN outcomes. Kissat stops intentionally in 30.628 seconds with 59,061 trace bytes (SHA-256 prefix a530b9cc); Tactix stops terminally in 94.177 seconds with 59,713 bytes (prefix 68cc8d74). The pre-bump raw heaps already differ, f18f4c4f7262a867 versus 29fc4824782cd181. The four common bumps preserve that split and end at the known hashes; there are no conflict-1,007 heap reinserts. Conflict 1,006 ends with an exact heap at decision count 23,744, while conflict 1,007 analysis begins at 23,786. Resume by hashing after each of those 42 common intervening decisions; the first unequal decision/lazy-pop endpoint is causal.

That decision sequence is campaign

```
20260819-1353-tierd-rank22-c1007-decision-heaps
```

It opens detail at conflict 1,006, starts decision heap hashing at count 23,745, and stops at conflict 1,007. Both immutable tasks use two CPUs and 16 GiB independently, with no throttle or node constraint. Scheduler dry validation passed as job 1634690; the actual pair is Slurm job 1634691. Resume by waiting for both arms, validating answer/trace/cleanup integrity, then pairing each common decision with its following hash and selecting the first mismatch.

Both job 1634691 tasks are valid, untruncated, and residue-free UNKNOWN outcomes. Kissat stops intentionally in 54.935 seconds with 5,017,839 trace bytes (SHA-256 prefix b98bdd3c); Tactix stops terminally in 115.161 seconds with 4,943,538 bytes (prefix e448f6ea). All 42 decision literals are identical. The raw hashes cannot yet support a causal claim: the presumed exact conflict-1,006 Kissat checkpoint came from older pin 682a4e7c, whereas this run uses 8ce35871. At decision 23,745 it publishes 56e0c19070e1f1b3; Tactix publishes both the unchanged pre-cleanup cb7647e239c08a03 and post-cleanup da71039987aa00be, reflecting different event timing. Resume by building one fresh current Kissat observer from the frozen source and current overlay, then use only that pin for the conflict-1,004–1,007 endpoint/pre-bump sequence before attributing a mutation.

A fresh current Kissat observer was built from frozen source 8af8e56f174b and the current overlay. The optimized build and static overlay test pass; its immutable SHA-256 prefix is 9a2b8a88. The same-pin endpoint replay is campaign

```
20260819-1400-tierd-rank22-current-pin-endpoints
```

at conflicts 1,004–1,007. Eight immutable tasks use two CPUs and 16 GiB independently, with no throttle or node constraint. Scheduler dry validation passed as job 1634693; the actual array is Slurm job 1634694. Resume by waiting for every task, validating answer/trace/cleanup integrity, and identifying the first unequal endpoint without cross-pin inference.

All eight job 1634694 tasks are valid intentional-stop UNKNOWNs with untruncated traces, deleted artifacts, and no pending cleanup. With the fresh current Kissat pin, raw heap hashes match exactly at conflicts 1,004, 1,005, and 1,006: c347670f0d35b6c8, f630e2a89e515c73, and cb7647e239c08a03. They first differ at conflict 1,007: 2dc7247556f5f30b versus d059788fa9995d69. This reproduces the frontier without cross-pin inference. The continuous trace places the mutation after conflict-1,006 analysis and before decision 23,745, during propagation of the asserting assignment. Resume by tracing activation/general heap-push operations in that interval; score bumps and backtrack reinserts are already exonerated.

The next diagnostic observer is now implemented at the central score-heap insertion primitive in both solvers. It records the inserted external variable, exact score, append position, and final bubbled position only while stable detail tracing is active; production builds erase the hook. Release passes 72/72 tests, Trace passes 71/71, and the Kissat overlay static test passes. Resume by building a fresh immutable Kissat observer from this overlay, then run one independent two-CPU/16-GiB pair

through conflict 1,007 and compare only insertions after conflict-1,006 analysis and before decision 23,745. Rank 22 remains open; this is localization evidence, not a repair.

The fresh heap-insertion observer pins are Kissat SHA-256 prefix `8ae2afa2` and Tactix prefix `24c47384`. The immutable campaign is

```
20260819-1412-tierd-rank22-c1006-heap-inserts
```

Its two tasks independently use two CPUs and 16 GiB, with no array throttle or node constraint; detail is restricted to conflict 1,006 and execution stops at conflict 1,007. Scheduler dry validation passed as job `1634702`; the actual array is Slurm job `1634703`. Resume by waiting for both tasks, requiring a valid answer, untruncated trace, deleted scratch, and no pending cleanup, then align the ordered `heap_insert` payloads after conflict-1,006 analysis and before decision 23,745.

Job `1634703` exonerates the central heap-insertion stream but does not produce two valid terminal arms. Tactix stops intentionally in 97.468 seconds with 46,331 trace bytes (SHA-256 prefix `3b88a2b6`); Kissat reaches all 42 decisions through 23,786, then unrestricted retained Sweep detail fills essentially the 100,000-record cap and the task times out at 180.137 seconds with 14,475,264 bytes (prefix `4d6092d3`). Both tasks delete scratch and leave no cleanup pending. Before that cap, both contain the complete 140-event score-heap insertion stream before conflict-1,006 analysis ends. Every payload matches in order, with common sequence SHA-256 prefix `2efcc434`; neither solver inserts between that analysis endpoint and decision 23,745. Treat only this completed target stream as diagnostic evidence, not the Kissat terminal arm. Resume with heap-pop-only detail for the first decision, using decision counter 23,745 in Kissat and 23,744 in Tactix because the solvers increment on opposite sides of variable selection.

That counter-aligned pop comparison is campaign

```
20260819-1419-tierd-rank22-d23745-heap-pop
```

It reuses pins `8ae2afa2`/`24c47384`, retains only heap-pop, heap-score, decision, and analysis-end records, and stops at conflict 1,007. The two tasks independently use two CPUs and 16 GiB with no throttle or node constraint. Scheduler dry validation passed as job `1634705`; the actual pair is Slurm job `1634706`. Resume by waiting for both tasks, validating answer/trace/cleanup integrity, then aligning the begin/move/end payloads for the logical decision 23,745 selection.

Job `1634706` is invalid as a cross-solver comparison. Both tasks delete scratch and leave no cleanup pending. Tactix reaches the intended stop in 97.811 seconds with 6,332 bytes (SHA-256 prefix `31acecc4`) and records no lazy-pop event for the first logical decision. Kissat times out at 180.068 seconds and leaves an empty buffered trace. The pop-only filter drops `heap_insert` records only after their payload has been formatted, so the millions of filtered calls still dominate the Kissat arm; its stream is also not line-buffered. Do not infer pop parity from this pair. The remaining source-backed candidate is arbitrary heap removal during root-variable deactivation, implemented by Kissat's `pop_heap` and Tactix's `heap_remove`, not the observed lazy pop-max path. Resume by making filtered insertions cheap, line-buffering Kissat, and tracing those arbitrary removals at conflict 1,006/decision count 23,744.

The arbitrary-removal observer is now implemented symmetrically at the two central heap primitives. It records removed and replacement variables with exact scores, the removed slot, replacement source slot, and final bubbled slot, restricted by an exact decision counter. The pop-only mode now rejects insertions before payload formatting, and Kissat is line-buffered. Release passes 72/72 tests, Trace passes 71/71, the overlay static test passes, and a fresh optimized Kissat overlay build passes. The fresh immutable SHA-256 prefixes are Kissat `fac07c1f` and Tactix `d7eaf69c`. Resume by pinning these binaries and tracing conflict 1,006 at decision count 23,744.

The immutable removal campaign is

`20260819-1431-tierd-rank22-c1006-heap-removes`

It restricts removal records to conflict 1,006/decision count 23,744 and stops at conflict 1,007. Two tasks independently use two CPUs and 16 GiB, with no array throttle or node constraint. Scheduler dry validation passed as job `1634708`; the actual pair is Slurm job `1634709`. Resume by waiting for both tasks, requiring valid answers, untruncated line-buffered traces, deleted scratch, and no pending cleanup, then align the ordered removal payloads from the last exact heap checkpoint.

Job `1634500` is another valid, untruncated, residue-free UNKNOWN timeout: 180.117 seconds, 24,576 bytes, SHA-256 prefix `9a8cdd49`, and conflict 70 only. Resume with one isolated 292-second cutoff, below the observed 296.719-second traced solve, and still require actual conflict-1,000 coverage.

The 292-second isolated continuation is campaign

`20260819-0720-tierd-rank22-kissat-c1000-heap292`

and Slurm job `1634501`. Resume by requiring conflict-1,000 coverage and comparing with saved Tactix task 1 from `1634485`.

Job `1634501` is clean but still incomplete: valid UNKNOWN, 292.118 seconds, an untruncated 24,576-byte trace with SHA-256 prefix `9a8cdd49`, zero residue, and conflict 70 only. Timeout-only capture is therefore unreliable. Resume with the disabled-by-default Kissat observer control that flushes and terminates after a selected conflict's analysis; run long enough to cross the c70 operation and stop precisely after conflict 1,000.

The stop control is committed as `d465475`; fresh Kissat observer SHA-256 prefix `9b5ee8fc` builds and passes the overlay regression. Campaign

`20260819-0728-tierd-rank22-kissat-c1000-stop`

runs as Slurm job `1634502`, isolated with a 600-second ceiling and an explicit flush/stop after conflict-1,000 analysis. Resume by requiring the analysis-end and heap digest, then compare saved Tactix task 1.

Job `1634502` completes as a valid, residue-free UNKNOWN timeout after 600.117 seconds. Its 32,768-byte trace is untruncated (SHA-256 prefix `2704cd67`) but reaches only conflict 584, so neither the conflict-1,000 analysis-end nor its digest exists. This isolates a diagnostic-observer cost: the detail window already suppresses pre-target heap hashes, but every macro snapshot still scans roughly 3.9 million variables and stable top-32 selection repeats that scan. Endpoint-only stop mode now suppresses pre-target macro snapshots and intermediate heap hashes, emits one final heap digest immediately before the selected analysis-end, then stops. Its static regression and fresh O3 observer build pass; binary SHA-256 prefix is `682a4e7c`. Resume by rerunning one isolated conflict-1,000 arm and comparing its endpoint digest with saved Tactix task 1 from job `1634485`.

The endpoint-only continuation is campaign

`20260819-0745-tierd-rank22-kissat-c1000-endpoint`

and Slurm job `1634505`. It runs one immutable Kissat observer pin with 16 GiB, two CPUs, a 600-second solver ceiling, no array throttle, and no node constraint. Resume by requiring exactly one conflict-1,000 `heap_scores` record immediately before `analysis_end`, checking an untruncated trace and zero cleanup residue, then comparing its ordered heap hash with saved Tactix task 1 from job `1634485`.

Limitations and resume instructions

The next shadow arm forces RUP replay on value-derived units. Exact population tracking has excluded unbacked deletion and missing direct-Y1 premises. The complete evidence is under `benchmarks/runs/`, in campaign

`20260818-2136-ascon-drat-input-restatement`

Do not rerun the complete 28B+5C or ranks 21–40 arrays until a focused proof verifies. If those later gates pass, resume divergence analysis at the earliest portable mismatch of the next still-open frozen rank. Ranks 22, 24, 26, 27, 30–37, 39, and 40 remain open. The 16 GiB/1,200-second results are iteration evidence, not replacements for the canonical 32 GiB/5,000-second gates.